Position Sensorless Control of a BLDC Motor Drive Using a Sliding Mode Observer

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MASTER OF TECHNOLOGY

by

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Abstract

In recent years, brushless DC (BLDC) motors have gained significant popularity due to their high efficiency, reliability, and low maintenance requirements. In traditional BLDC drives, the rotor position is sensed using Hall-effect sensors, which allow the windings to be switched at the appropriate times. However, such sensors make the hardware more complex and expensive, and they also require additional wiring. Sensorless control, on the other hand, estimates the rotor position and speed from electrical variables.

A sensorless BLDC motor drive based on a sliding mode observer (SMO) is proposed in this paper. The observer estimates the back EMF, rotor angle, and speed. To obtain smooth switching action and reduce the chattering associated with the traditional signum function, a sigmoid function is used as the nonlinearity in the SMO. The estimated speed is fed to a PI controller that produces the PWM signal, while the estimated rotor angle is used in the six-step commutation algorithm, which generates the gate pulses for the inverter.

The complete control system was implemented in a MATLAB/Simulink model, and its performance was analyzed using the back EMF, rotor position, rotor speed, and other related waveforms.

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List of Abbreviations

BLDC	Brushless DC Motor
EMF	Electromotive Force
SMO	Sliding Mode Observer
PWM	Pulse-Width Modulation
FOC	Field-Oriented Control
PMSM	Permanent Magnet Synchronous Motor
VSC	Voltage Source Converter
IGBT	Insulated Gate Bipolar Transistor
CRO	Cathode Ray Oscilloscope

Chapter 1

Introduction

1.1 Introduction to BLDC Motors

Due to the increased efficiency, small size, good torque-to-weight ratio, and low maintenance required, BLDC motors have become widely used in today's electric drives. The BLDC motor has permanent magnets at the rotor, and the windings at the stator. In contrast to the traditional DC motor, the BLDC motor does not contain brushes or a commutator. The stator currents are switched electronically using the power inverter.

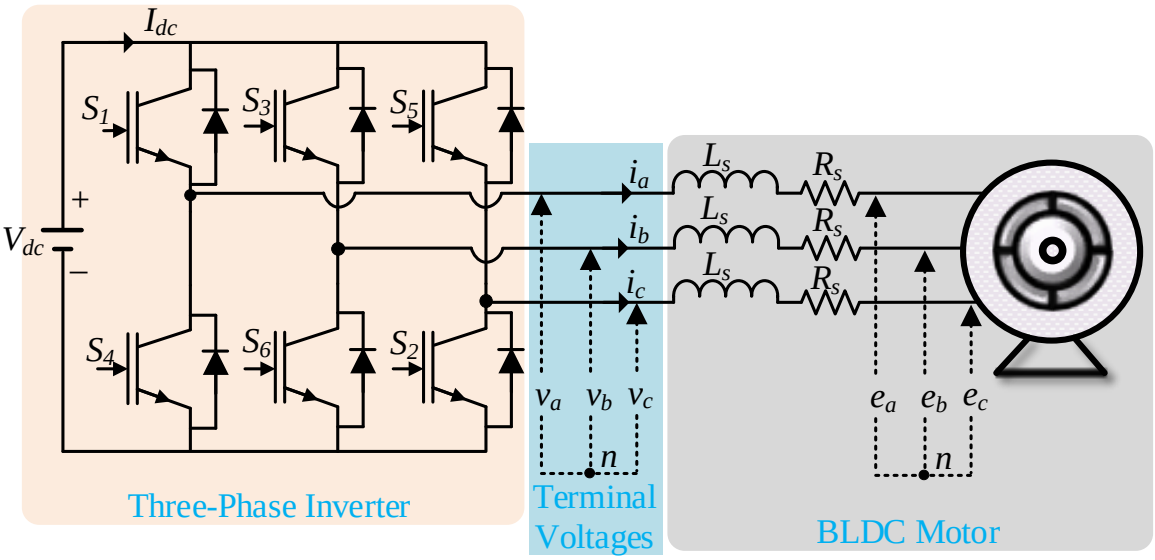


Figure 1.1: Structure of BLDC motor.

The operation of the motor occurs due to the interaction of the rotor permanent magnet field and the field created by the stator windings. Control of the stator current allows one to control the electromagnetic torque and the speed. The structure of the drive includes a DC power source, the three-phase inverter, the BLDC motor, and the controller to generate switching signals.

The knowledge of the rotor position is very important, since this information is used by the controller to switch on the appropriate winding. Traditionally, this was performed by means of the Hall-effect sensors, but they require additional wiring. Therefore, there were developed several different techniques which use estimation of the rotor position and speed using only electrical measurements.

Sensorless control involves back EMF, flux estimation, and observer methods. The role of the observer is to estimate the internal state of the motor based on its mathematical model and measured electrical signals, hence, the performance of such an observer will largely depend on the model accuracy and measurement. This means that sensorless control methods may be applied to estimate the speed and position of BLDC motors. [1], [2]. The detailed structure of the BLDC motor is shown in Fig. 1.1

1.1.1 Construction and Operating Principle

BLDC motor basically consists of stator, rotor with permanent magnets, shaft, bearings, and electronic drive circuit. Stator comprises three-phase windings which are located in the slots of laminated magnetic core. The permanent magnets mounted in the rotor generate the required magnetic field. These permanent magnets can be either installed on the rotor or located inside the rotor depending on the motor construction.

Stator windings are connected to the DC supply source through the three-phase voltage source inverter. Inverter contains six controllable switches arranged in three legs, and switching state is determined by the rotor position and mode of operation.

In conventional six step commutation method, only two phases conduct at any time and third one remains open. Commutation occurs after each 60 degrees electrical.

Relation between the electrical and mechanical speeds is:

$$\omega_e = p\omega_m$$

where ω_e is electrical angular speed, ω_m represents the mechanical angular speed, and p gives number of rotor pole pairs.

Without brushes, there is no mechanical commutation and hence less wear related to brush contacts. Therefore, BLDC motors are applicable in such areas as electric cars, industry drives, cooling, robots, and automatic machines. However, their performance depends on proper commutation and control of currents. [3].

1.1.2 Equivalent Circuit of a BLDC Motor

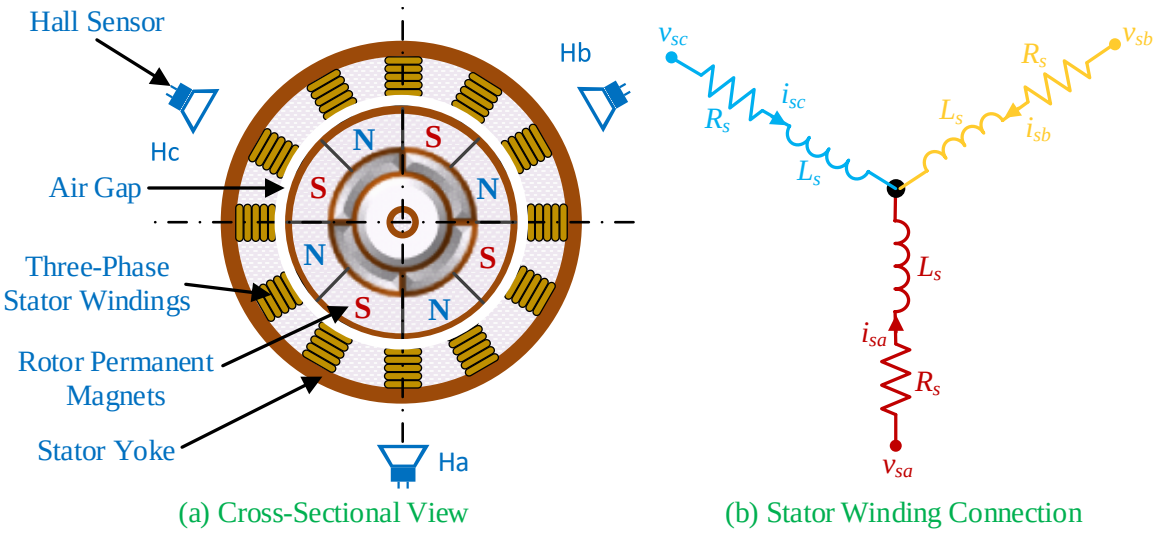


Figure 1.2: Equivalent circuit of BLDC motor.

The equivalent circuit of a BLDC motor represents the electrical behaviour of the stator windings. Each phase can be modelled using a winding resistance, an effective inductance, and a back-EMF source as shown in Fig. 1.2. The applied phase voltage must overcome the resistive voltage drop, the inductive voltage drop, and the induced back-EMF.

For a balanced three-phase motor, the phase-voltage equations can be written as

$$v_a = R_s i_a + L_s \frac{di_a}{dt} + e_a$$

$$v_b = R_s i_b + L_s \frac{di_b}{dt} + e_b$$

$$v_c = R_s i_c + L_s \frac{di_c}{dt} + e_c$$

where v_a, v_b, v_c represent the phase voltages, i_a, i_b, i_c represent the phase currents, R_s is the phase resistance, L_s is the effective phase inductance, and e_a, e_b, e_c are the phase back EMFs.

These simplified equations assume that mutual coupling is either neglected or incorporated into the effective inductance. If mutual inductance is explicitly included, the inductive voltage terms must be modified accordingly.

For a three-wire stator without a neutral connection,

$$i_a + i_b + i_c = 0$$

The electromagnetic power converted by the motor is

$$P_e = e_a i_a + e_b i_b + e_c i_c$$

and the developed electromagnetic torque can be expressed as

$$T_e = \frac{e_a i_a + e_b i_b + e_c i_c}{\omega_m}$$

for nonzero mechanical speed, with consistent phase and speed conventions.

The equivalent circuit provides the foundation for developing the motor's mathematical model and analysing the relationship between phase voltage, current, back-EMF, and electromagnetic torque. It is also useful in designing model-based observers, which rely on electrical equations to estimate unmeasured motor states [4].

1.1.3 Back Electromotive Force Characteristics

As the rotor rotates, there will be relative motion between the magnetic field of the rotor and the conductors of the stator, causing an induced voltage in the windings of the stator. This induced voltage is called the back EMF. The magnitude of the back EMF is determined by two factors: the speed of the rotor and the flux linkage of windings.

The phase back EMFs can be represented by

$$e_a = K_e \omega_m f(\theta_e)$$

$$e_b = K_e \omega_m f(\theta_e - 120^\circ)$$

$$e_c = K_e \omega_m f(\theta_e - 240^\circ)$$

where K_e represents back EMF constant, θ_e is the electrical position of rotor, and $f(\theta_e)$ represents the normalized waveform.

For a typical BLDC motor, the back-EMF of phases will take an approximate trapezoidal form. All three phases are staggered by 120 electrical degrees. The exact form of the back-EMF is dependent upon rotor magnetic pole configuration, air-gap flux density, and stator windings configuration.

Back-EMF is crucial for both the generation of torque and sensorless control.

As the drive system is a sensorless one, back-EMF data can be used to determine rotor state information and rotor speed. But when the drive system is at standstill, the back-EMF induced by rotation will be zero. This makes accurate rotor-state estimation challenging during startup and low speed operation.

1.1.4 Mathematical Modelling of a BLDC Motor

Mathematical modeling is required for analyzing its dynamic behavior and for designing the necessary control algorithms. This model relates the electrical parameters with the electromagnetic torque, rotor velocity, and rotor position.

Electrical subsystem can be modeled by the phase voltage equations presented in Section 1.1.2. Electromagnetic torque is the product of phase current and corresponding back EMF:

$$T_e = \frac{e_a i_a + e_b i_b + e_c i_c}{\omega_m}$$

The mechanical dynamics follow from the rotational equation of motion:

$$J\frac{d\omega_m}{dt} = T_e - T_L - B\omega_m$$

where J is the combined inretia of rotor and load, T_L is the load torque, and B is the viscous friction coefficient.

The rotor angle theta is estimated through integration of mechanical speed:

$$\frac{d\theta_m}{dt} = \omega_m$$

The electrical position of rotor is related to the mechanical position by

$$\theta_e = p\theta_m$$

where p represents number of pole pairs.

The comprehensive motor model includes the electrical equations, back EMF equations, torque equation, and mechanical equations. In a modeling environment like MATLAB/Simulink, these equations can be used to study the response of the motor under differnt voltages and loads.

The mathematical model is also at the core of the observer-based sensorless control approach. The observer utilizes the motor model and measurement signals, for example, the stator current and voltage, to estimate the unmeasured states. The errors in the motor parameters, measurement noise, and non-modeled inverter dynamics

may affect the accuracy of the state estimation [5].

1.1.5 BLDC Motor Commutation Techniques

The method of switching the phase that is conducting of the stator in relation to position of the rotor such that the motor develops torque in the required direction is known as commutation. The methods of commutation in a BLDC drive are accomplished by the switching of the inverter components. The main commutation techniques are explained below [6].

Hall-sensor-based commutation: Hall-effect sensors detect changes in the rotor magnetic field and provides digital position signals. In a typical three-sensor configuration, the combined Hall states identify six rotor sectors per electrical cycle [7]. The controller uses a predefined switching table to select the inverter devices. This method is relatively simple and provides direct rotor-sector information, but it requires physical sensors and additional connections.

Back-EMF-based sensorless commutation: In conventional zero-crossing detection, the back EMF of the phase thatis not conducting is monitored. Its zero crossing provides a reference for determining the next commutation instant. For ideal six-step operation, commutation is commonly scheduled 30 electrical degrees after the zero crossing. Practical implementations must account for filtering delay, switching noise, and deviations from the ideal waveform [8].

Observer-based commutation: An observer estimates quantities such as back-EMF, rotor position, or speed from a mathematical model and measured electrical signals. The estimated information is then used to determine suitable commutation instants. Sliding-mode observers and other model-based estimation methods have been investigated for sensorless BLDC drives [9], [10].

Each method involves a trade-off between implementation complexity, estimation accuracy, cost, and operating-speed range. In particular, methods that depend directly on back-EMF encounter difficulties at low speed because the induced voltage decreases with rotational speed.

1.1.6 Comparison of Trapezoidal and Sinusoidal Back-EMF

The back EMF waveform plays an important role in choosing the current control technique of a permanent magnet motor. Conventional BLDC operation is commonly associated with trapezoidal back-EMF and six-step commutation, whereas sinusoidal permanent-magnet motor operation uses approximately sinusoidal back-EMF and si-

nusoidal phase currents.

For trapezoidal back-EMF, rectangular or quasi-rectangular phase currents are used to obtain relatively constant electromagnetic torque during the normal conduction interval. In sinusoidal operation, the phase currents are controlled to follow sinusoidal references, which can produce smooth torque when the current waveforms are appropriately aligned with the back-EMF. Characteristic Trapezoidal back-EMF Sinusoidal back-EMF Typical waveform Trapezoidal Sinusoidal Typical current control Six-step commutation Sinusoidal current control or field-oriented control Control implementation Relatively simple for conventional six-step drives Generally requires more elaborate current regulation Torque characteristics Ripple may occur during commutation Smooth torque is possible with accurate current regulation Common considerations Commutation timing and current transfer Current tracking, phase alignment, and controller tuning.

These are typical configurations rather than strict limitations. Actual torque ripple depends on the motor design, current waveform, inverter switching, and control implementation. A sinusoidal motor operated with six-step current control may exhibit additional torque ripple, while suitable current shaping can improve torque characteristics in a trapezoidal-back-EMF motor.

For sensorless operation, the waveform also affects how rotor position is estimated. Nonideal back-EMF, harmonic components, and switching disturbances may cause errors in position detection or observer output. Consequently, the estimation method should be selected according to the actual motor characteristics and required operating range [11] , [12].

1.2 Hall Sensor Based Control of BLDC Motors

The use of Hall effect sensors is an ordinary technique to obtain the rotor position for the BLDC drive. These sensors are used to detect the magnetic field generated by the rotor magnets and generate digital pulses that depends on the position of rotor. The typical three phase motor uses three sensors, which provide six positions over one electrical cycle.

The three Hall signals are generally represented as H_a, H_b, and H_c. Their combined states identify the rotor sector, which is used by the controller to select the conducting inverter switches. The switching sequence ensures that the stator phases are energized in the required order to maintain rotation.

The speed can also be controlled using feedback control. In this the measured speed is compared to the reference, and the difference is taken into the PI controller.

The output from the controller determines the PWM duty cycle for the inverter to adjust the voltage and current.

A typical control structure is

$$e_\omega = \omega_{\text{ref}} - \omega_m$$

$$u(t) = K_p e_\omega + K_i \int e_\omega \, dt$$

where ω_{ref} is the reference speed, ω_m is the measured speed, e_ω is the speed error, and K_p and K_i are the proportional and integral gains.

The main advantage of Hall based control is that it provides direct rotor-sector information, making commutation relatively straightforward. However, the physical sensors increase the number of connections and may be affected by installation constraints, temperature, vibration, or sensor faults. Their discrete position information also has limited angular resolution unless additional estimation or measurement techniques are employed.

These limitations motivate sensorless methods that obtain the required rotor information from the electrical behaviour of the motor instead of relying entirely on dedicated position sensors [13].

1.3 Position Sensorless Control of BLDC Motors

In sensorless position control, the rotor position sensor is not physically used in BLDC motor. In this method, the controller calculates the rotor position and velocity from the electrical variables like stator current, voltage and back-EMF. The use of position sensors can avoid the wiring problem and make the motor simpler.

Sensorless estimation can be generally classified into two types, back EMF based methods and model based estimation methods. In the case of back EMF based methods, rotor position is calculated on the basis of some parameters of the induced voltage, such as zero-crossing of the phase that is not conducting. In a model based approach, the observer receives the applied voltage and measured current as inputs. It uses these signals together with the mathematical model to estimate quantities that are not directly measured. A general observer structure can be represented as [14].

In a model-based approach, the observer receives the applied voltage and measured current as inputs. It uses these signals together with the mathematical model to estimate quantities that are not directly measured. A general observer structure

can be represented as

$$\dot{\hat{x}} = f(\hat{x}, u) + L(y - \hat{y})$$

$$\hat{y} = h(\hat{x})$$

where $\hat{x}$ is the estimated state vector, u represents the input signals, y is the measured output, $\hat{y}$ is the estimated output, and L represents the observer correction gain. The exact equations and gain structure depend on the selected observer.

The estimation error is defined as

$$e_x = x - \hat{x}$$

where x is the actual state and $\hat{x}$ is its estimate. The observer is designed so that this error remains bounded and, under the required operating conditions, converges towards zero or a sufficiently small neighbourhood of zero.

A major challenge in sensorless BLDC control is maintaining reliable estimation over a range of speed. At low speed, back EMF is weak, making position information more difficult to extract. At higher speeds, switching noise, parameter mismatch, and phase delay can impact the accuracy of estimates. The performance of an observer therefore depends on its mathematical formulation, gain selection, measurement quality, and operating conditions [3].

Sensorless control is particularly useful in applications where reducing the number of physical sensors is desirable. Nevertheless, successful implementation requires an estimation method that can provide sufficiently accurate rotor information for reliable commutation and speed regulation.

1.4 Research Motivation and Problem Statement

BLDC drive performance depends on the reliability of rotor position. In traditional sensor based drives, rotor position information is provided using Hall sensors that generate the needed position signals for commutation. Despite the advantages of this method, additional components and wiring are required. This might increase the complexity of installation and not necessarily be favorable for compact drive designs.

In sensorless drives, rotor position can be estimated from electrical variables of the motor itself. However, estimating the rotor's position is not easy due to the back-EMF magnitude being dependent on speed, electrical disturbances during switching,

and discrepancies between practical and nominal values of the motor's parameters. These conditions may lead to the occurrence of estimation errors and wrong commutation, resulting in torque ripples and current distortions.

Detecting back-EMF directly becomes hard at low speeds, while observers based on models are often sensitive to parameter mismatches, noise, and gain tuning. That is why research on the design of more robust observers and estimation techniques was conducted to provide estimation of rotor position and speed [3].

In the present work, the development of a sensorless control for the operation of a BLDC motor based on the estimation technique is presented. The aim is to achieve estimation of the desired motor states based on available electric signals and using the estimated data in order to operate the motor without measuring the rotor position.

The performance of the above mentioned technique can be examined through comparison between the estimated data and their actual counterparts in simulation. Performance measures include estimation error, convergence behavior, speed performance, and the quality of the commutation signals generated.

1.5 Research Objectives

The central objective of this study is the development and analysis of observer-based position sensorless control of BLDC motor in MATLAB/Simulink. The proposed approach estimates the motor's back EMF, rotor electrical position, and rotational speed from the available electrical signals.

Some objectives of this work are as follows:

1. Study the working principle, modelling and electric parameters of the three-phase BLDC motor.

2. Analyze the sensorless control approaches and difficulties related to back-EMF-based rotor position detection, especially at low speeds.

3. Development of the SMO algorithm for the stator currents and back EMF components estimation in stationary α-β reference frame.

4.Development of second order observer for estimating the rotor's electrical speed and position based on the obtained back EMF components.

5. Implement six-step electronic commutation and voltage control through PWM technique in MATLAB/Simulink using rotor position estimation after the starting phase.

6. Create the speed control system based on PI regulator to maintain the desired

motor speed in accordance with the reference signal.

7. Test the performance of the given control strategy by comparing the back EMF estimation, rotor position, and speed with the real simulation data.

8. Study the behavior of the sensorless drive under the variation of operation conditions, such as load torque disturbance, and find out the limitations of the designed approach.

1.6 Scope of the Present Work

This paper is presented around the design and evalution of the observer based sensorless control approach for a three phase BLDC motor using computer simulations. The motor model, inverter, observer, commutation algorithm, PWM generator, and speed controller are modeled in MATLAB/Simulink environment.

A mathematical model of the motor will be employed in order to establish the correlation between the applied stator voltages, stator currents, and back-EMF. A sliding mode observer will estimate the current and generate back EMF signals. Before these signals are given into the second order observer, they will be passed through a low-pass filter.

The second-order observer uses the back EMF signal to calculate the estimated electrical speed and position. The estimated electrical speed is transformed into the mechanical speed which is utilized by speed control loop and compared with the desired speed. The rotor position estimated will be used to determine the proper commutation steps. A PWM strategy is implemented for controlling the average voltage while the speed error is controlled using a PI controller for the duty cycle.

Since back EMF signal is either not strong enough or missing in case of standstill conditions, the rotor position estimation process becomes very challenging. Thus, the current realization of the algorithm utilizes the real rotor position to initialize the commutation steps and then switches to the estimated position when the observer gains enough experience to provide meaningful results. In other words, the system operates in a sensorless mode only after the startup process ends.

The performance of the developed algorithm is analyzed via simulation waveforms. The estimated and measured components of the back-EMF signal, rotor position and speed are compared. The response of the closed-loop drive is also examined under load torque and speed disturbances. The study is limited to the simulated motor model and the operating conditions considered; experimental validation using a physical motor drive is outside the present scope.

Chapter 2

Literature Review

2.1 Introduction

There are several approach that have been developed to control the BLDC motors using information about the rotor position. The information may be acquired either through measurement or estimation. The sensor based method is the one that makes use of sensors to calculate the rotor position. The sensors feed back the position information of rotor in real time. In the sensorless method there is no use of any position sensor. This method estimates the rotor position through information of electrical characteristics such as voltage and currents of the motor [2, 4].

2.2 Review of Sensor-Based BLDC Motor Control

Hall sensors are commonly used to obtain position information of rotor for electronic commutation. Gamazo-Real et al. [2] discussed different approaches to speed and position control of BLDC motors, including sensor based as well as sensorless techniques. The Hall sensors serve as a simple way to detect the position of the rotor and find out its commutation sectors. Nevertheless, using Hall sensors demands an extra hardware installation. This is why sensorless approaches to BLDC motor control are considered.

2.3 Review of Sensorless BLDC Motor Control

For the sensorless systems, rotor position and speed are estimated through measurement. of electrical variables instead of having an additional position sensor. Various techniques have been explored for the purpose of determining rotor position. Çetintaş

et al.[4] studied position sensorless speed control of a BLDC motor using the back EMF method. Darba et al. [11] investigated sensorless commutation and speed control on the basis of back EMF symmetric threshold tracking. Fakham et al. [3] presented the design and implementation of a back EMF sliding mode observer for a BLDC motor. These studies address different aspects of sensorless operation, including back-EMF detection, commutation and observer-based estimation.

2.3.1 Back-EMF-Based Sensorless Techniques

Back EMF based techniques use the induced voltage in the windings of stator to obtain information about position of rotor. In six-step operation, the zero crossing of the back EMF in the phase that is not conducting can be used to obtain the appropriate instant of commutation. Çetintaş et al. [4] investigated position sensorless speed control using the back-EMF method. Darba et al. [11] studied a symmetric threshold tracking technique for sensorless commutation and speed control. Soni and Tripathi [13] proposed a back EMF noise band based initial startup scheme for sensorless BLDC operation. Zhang et al. [14] investigated commutation deviation correction in the presence of asymmetric back-EMF waveforms. These studies tells the importance of back EMF information for rotor and position estimation and accurate commutation.

2.3.2 Terminal-Voltage-Based Sensorless Techniques

Terminal-voltage-based techniques obtain information about rotor position through electrical voltage measurements at the motor terminals. In conventional six-step operation, the phase terminal voltage that is not conducting can be analysed to identify the zero-crossing point of the back EMF. The detected position information is used to determine the commutation instant. Çetintaş et al. [4] investigated a back EMF based position sensorless method for BLDC motor speed control, while Darba et al. [11] studied symmetric threshold tracking for sensorless commutation. These approaches tells the importance of voltage based position detection in sensorless BLDC motor operation.

2.3.3 Observer-Based Sensorless Techniques

Observer based methods estimate motor states using a mathematical model and available electrical measurements. Based on the estimation method, the estimated quantities may include back EMF, position and speed of rotor. Fakham et al. [3] presented a back EMF sliding mode observer for a BLDC motor and investigated its practical

implementation. Fakham et al. [5] studied sliding mode observer based estimation of position and speed of rotor in a BLDC motor. Deenadayalan et al. [6] investigated a modified sliding mode observer for wide speed range operation.

2.3.4 Sliding Mode Observer-Based Techniques

A SMO uses a correction term to drive the estimation error towards a predefined sliding surface. Qiao et al. [1] proposed a sliding mode observer for position sensorless control of a PMSM. Their approach provides a reference for estimating position and speed of rotor using observer based techniques.

Several studies have investigated sliding-mode observers specifically for BLDC motor applications. Fakham et al. [3] presented the design and implementation of a back EM -based sliding mode observer for a BLDC motor. Fakham et al. [5] investigated the estimation of rotor position and speed using a sliding mode observer. Deenadayalan et al. [6] studied a modified sliding mode observer for operation in a range of speed. Qiu et al. [9] studied position sensorless control of a BLDC motor based on a sliding-mode observer. In addition, Zhang et al. [10] investigated a variable reaching law based sliding mode back EMF observer for BLDC motor position estimation.

Although Qiao et al. considered a PMSM rather than a BLDC motor, their work provides a useful foundation for understanding sliding mode observer design. However, the observer performance must be evaluated using electrical model and operating characteristics of the BLDC motor. The estimation accuracy, convergence and response under different operating conditions are important aspects to investigate.

2.4 Research Gaps

The reviewed studies investigate sensorless control from different perspectives. Çetintaş et al. [4] and Darba et al. [11] examined back-EMF-based position detection and sensorless commutation. Fakham et al. [3] and Deenadayalan et al. [6] investigated sliding-mode observer-based estimation for BLDC motors. Gan et al. [7] and Zhang et al. [10] explored modifications to the sliding-mode observer structure. Qiao et al. [1] provide an important reference for observer design, although their investigation focuses on a PMSM.

The comparison between the studies suggests that the performance of the observer relies on the motor model, the estimation approach and operating conditions. The approach which has been designed to work with a certain kind of motor should

be further studied before being implemented with other types of motors. This work pays attention to implementation of the observer using a BLDC motor model in MATLAB/Simulink.

2.5 Motivation for the Proposed Approach

The reviewed studies provide a basis for investigating observer-based sensorless control of BLDC motors. The present work focuses on implementing a sliding-mode observer and examining its ability to estimate the motor states using the available electrical signals. The observer approach proposed by Qiao et al. [1] provides the basis for the selected methodology, while BLDC-specific observer studies as those by Fakham et al. [3] and Deenadayalan et al. [6] provide additional background.

Comparison between the actual and estimated signals allows estimating the estimation error as well as the transient and settling behavior in various operation regimes. It allows us to conclude about the applicability of the designed observer in case of the proposed sensorless control system.

2.6 Summary

This chapter considered the approaches for controlling the BLDC motors on the basis of physical position sensors and estimation methods. Back-EMF detection, position estimation based on terminal voltage and observer-based approaches were discussed. The previous works have shown the implementation of various estimation methods (extended Kalman filter and sliding mode observer) as well as the significance of commutation information, back-EMF estimation and observer design.

Chapter 3

Simulation of Hall Sensor Based Control of BLDC Motor

3.1 Introduction

This chapter explains the modeling process of a BLDC motor drive with Hall-sensors in MATLAB/Simulink. The model involves the following components: the DC source, three-phase inverter, BLDC motor, Hall signal, and the switching scheme involved in commutation. Both the electric and the mechanical aspects of the BLDC motor are modeled.

The idea of this chapter is to develop a controller based on Hall sensors as the reference system for comparison with the observer based sensorless controller of the next chapter.

3.2 BLDC Motor Drive System Configuration

A BLDC motor drive system consists of the following components: DC power supply, three-phase voltage source inverter, BLDC motor, Hall sensors, and a commutation logic circuit. The electrical energy from the DC source is fed to the inverter where the energy is transformed to three phase regulated voltages.

The three-phase inverter has six power switches that are configured in three legs of an inverter. Through control of the switches' operation, the current can be fed to the motor phases. With six-step commutation technique, only two motor phases carry current at any given time and third phase is left open. The phase to which current is not conducted depends on the switching configuration of the inverter and the flow of the current.

Hall sensors supply the position of the rotor through digital signals that have a total of three combinations. These combinations help determine rotor sector and switch-state for commutation.

3.3 MATLAB/Simulink Model

3.3.1 Simulated Circuit Diagram

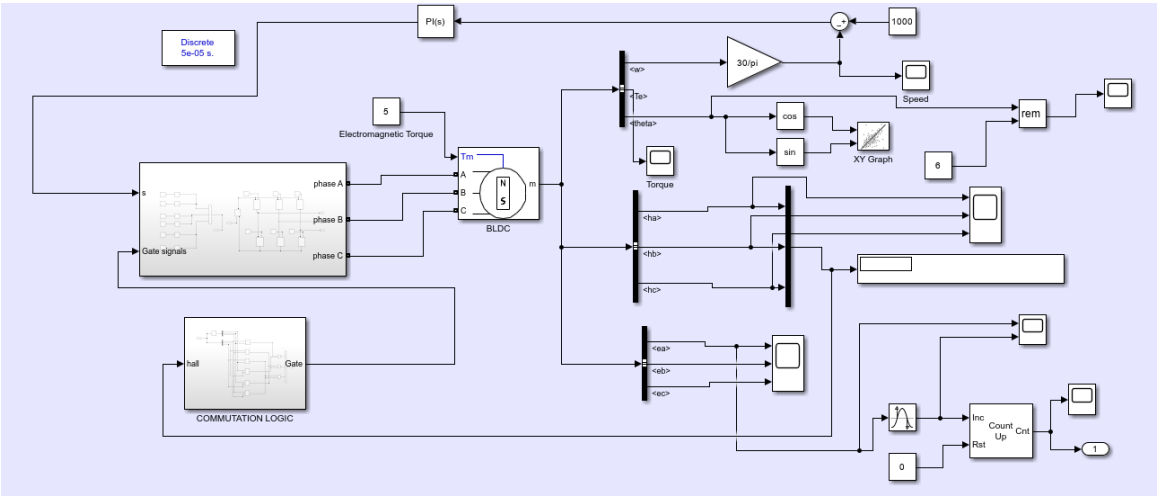


Figure 3.1: Simulated Circuit of Hall sensor based BLDC motor drive

3.3.2 Three-Phase Inverter Subsystem

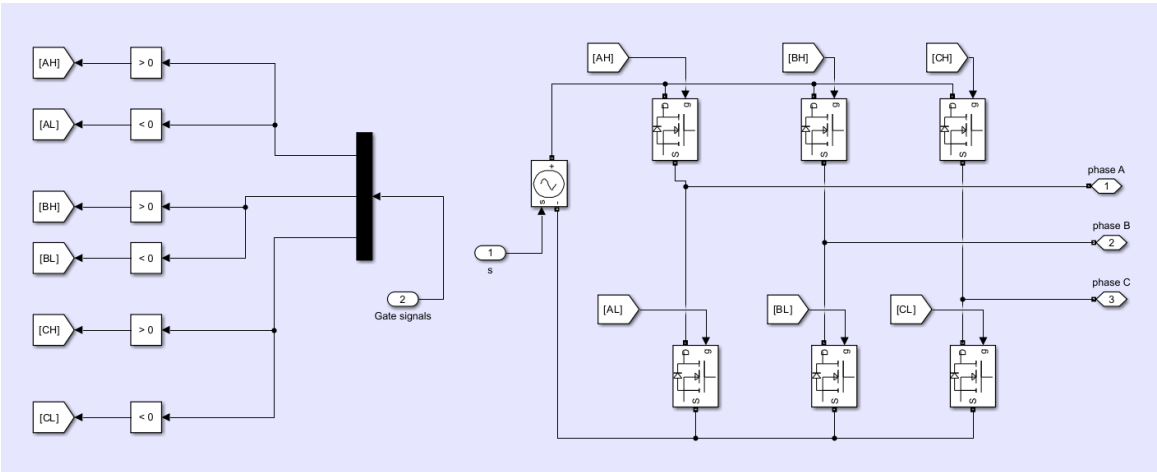


Figure 3.2: Three Phase Inverter Circuit

3.3.3 Commutation Signal Logic Subsystem

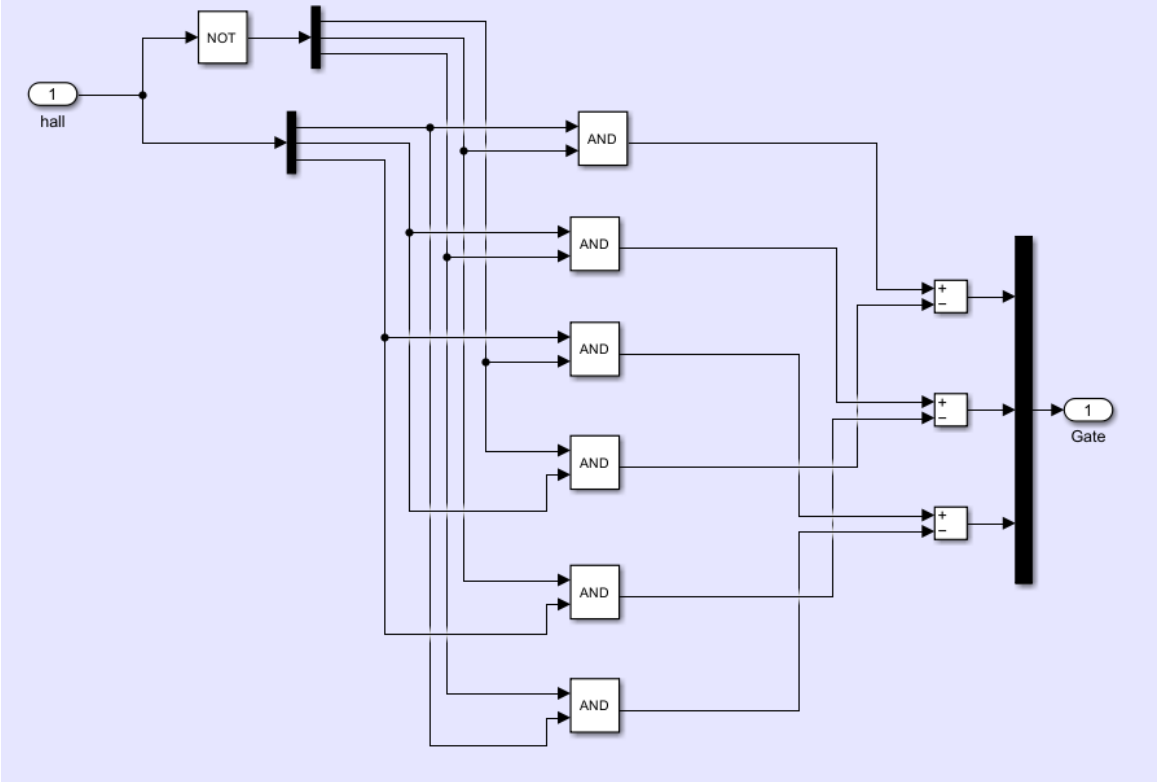


Figure 3.3: Hall-sensor-based commutation logic for inverter switching

3.4 Simulation Results

3.4.1 Rotor Position

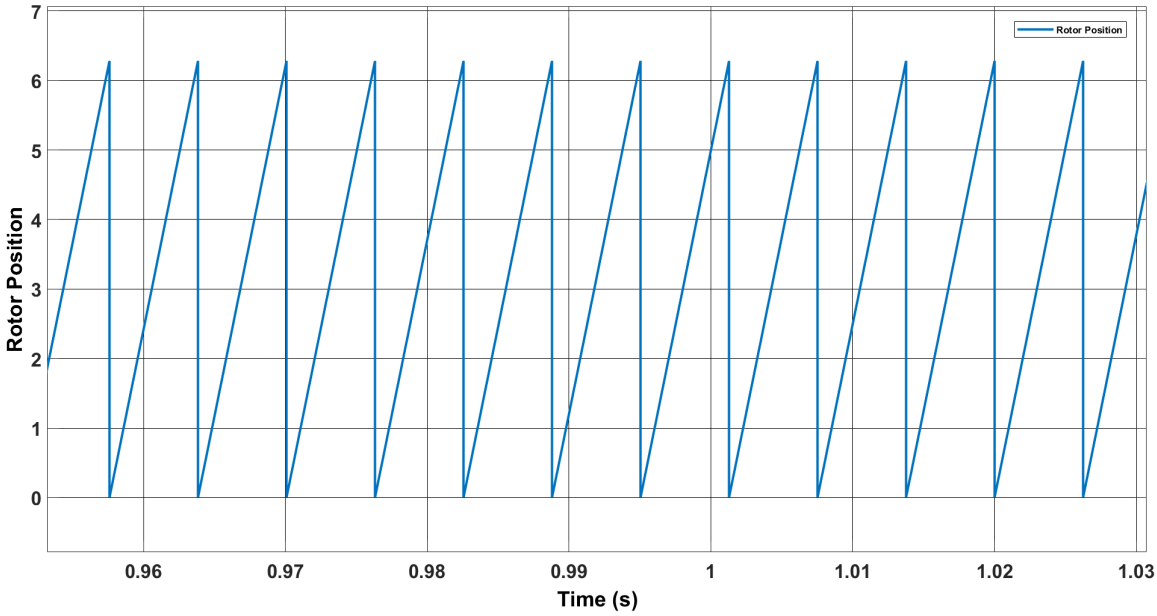


Figure 3.4: Rotor angular position

3.4.2 Rotor Speed (Reference Speed = 800 rpm)

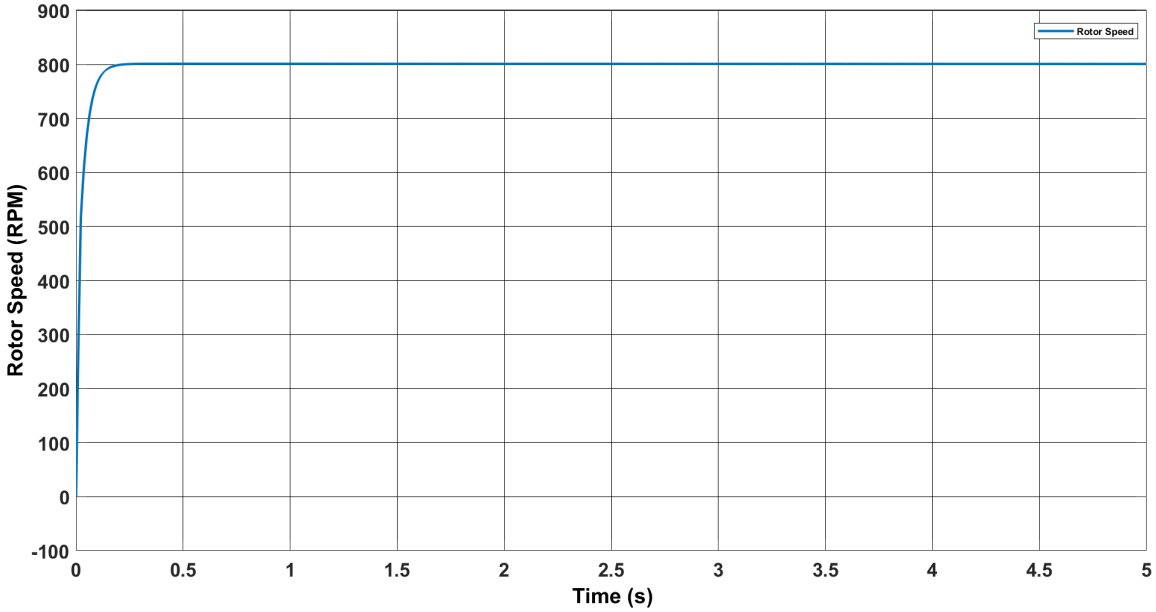


Figure 3.5: Rotor speed

3.4.3 Hall Signals

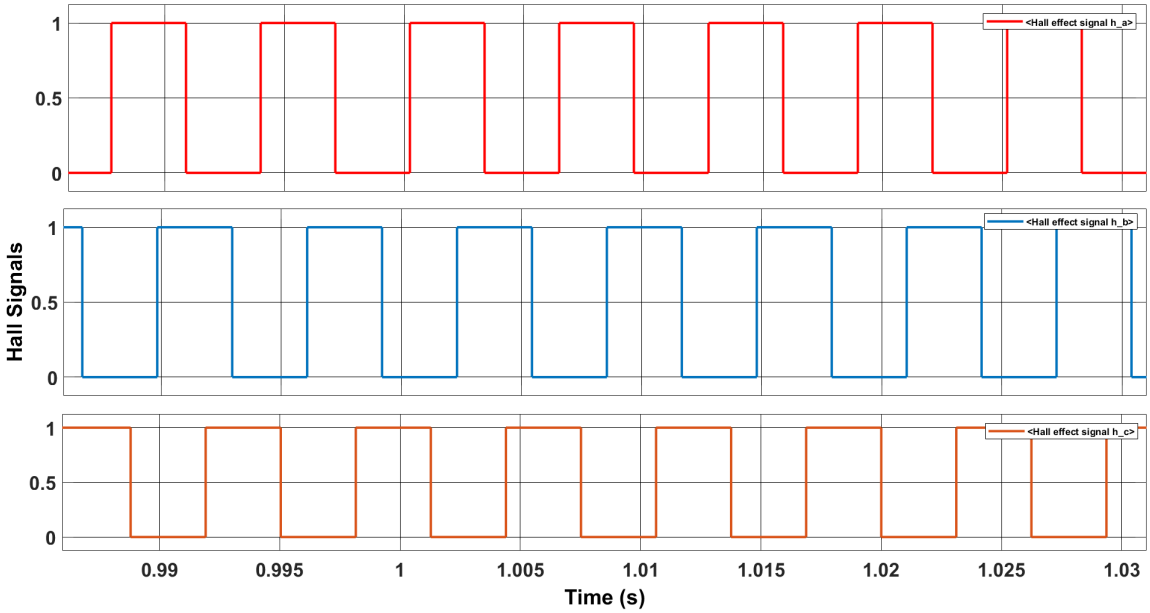


Figure 3.6: Hall signals

3.4.4 Back-EMF Waveform

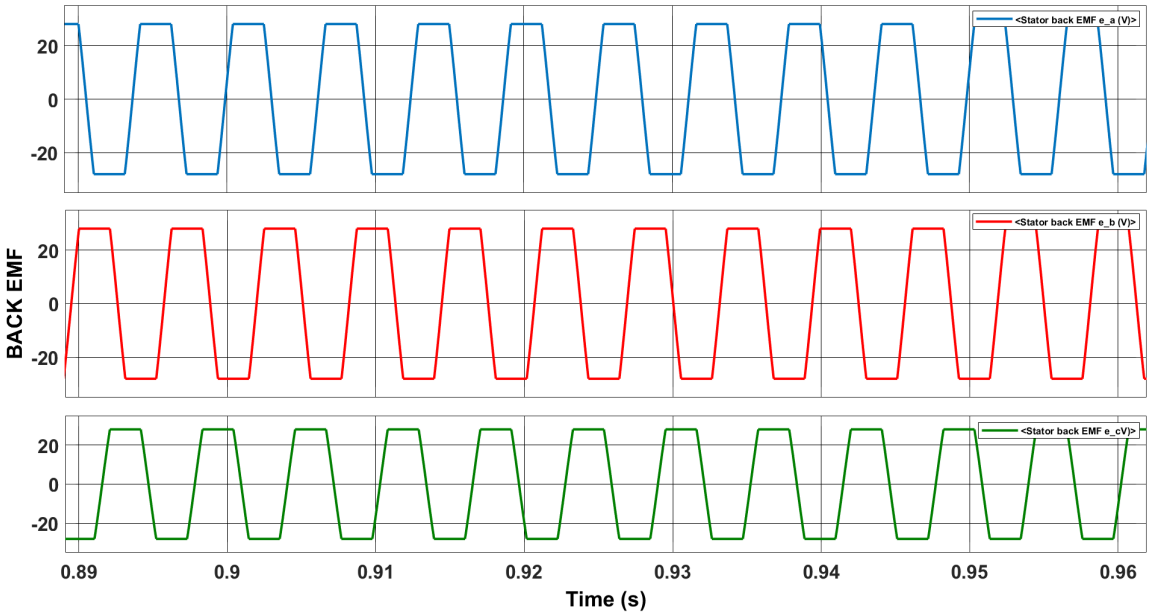


Figure 3.7: Back EMF Waveforms

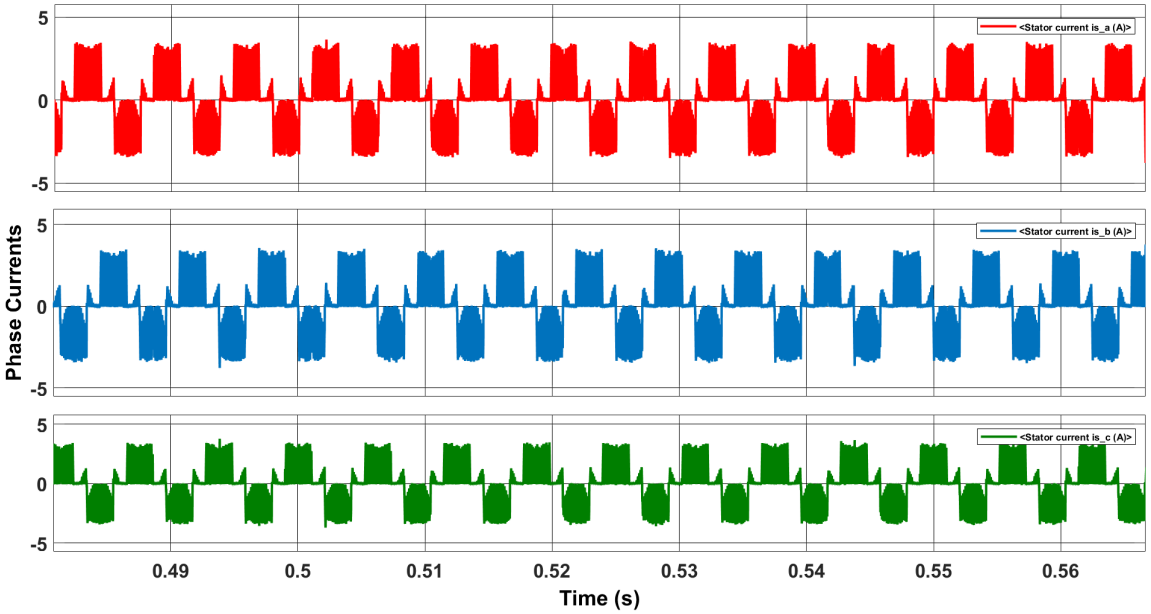


Figure 3.8: Phase Currents

3.5 Summary

BLDC Motor Hall Sensor Based Control Simulation proves to be an effective demonstration of the electronic commutation technique in terms of rotor position feedback. The three Hall sensors placed in 120° are used by the controller to detect rotor position and generate the correct sequence of switches to turn on/off the inverter switches. It can be seen from the simulations that the Hall sensor-based technique is a reliable and accurate technique to provide commutation, especially at low to medium speed. This technique is a simple, effective, and low-cost technique to use as a BLDC motor control method; thus, it is useful for electric cars, fans, and other industrial motors.

Chapter 4

Sensorless BLDC Motor Control Using a SMO

4.1 Introduction

BLDC motors are commonly used in drives because they are very efficient, small and need little maintenance. Conventional BLDC drives usually use Hall sensors to find out where the rotor is and to create the commutation sequence. These sensors add hardware and extra electrical connections.

Sensorless control figures out the rotor data from the motor's variables so it does not need separate rotor-position sensors. A common way to do this is by estimating back EMF. Back EMF based methods work well when the motor speed changes, when there is measurement noise, when motor parameters vary or when the estimate is delayed [2]. This paper considers an SMO design using the sigmoid function as the switching function for the sensorless control of a BLDC motor. Observer is implemented in MATLAB/Simulink based on the motor model and the estimated state variables. The results of the back EMF estimation process are used for calculation of the position and speed of rotor, which are then applied for commutation of a three phase inverter.

This chapter's goal is to create the sigmoid-based observer, implement the estimation process in MATLAB/Simulink and test its performance, with BLDC motor signals. The results are then compared with those of the Hall sensor based BLDC drive.

4.2 Proposed Sensorless Drive Configuration

The proposed sensorless drive system consists of five main components, which include the DC source, three-phase voltage-source inverter, brushless DC motor, signal transformation unit, sliding mode observer, and commutation signal generator. The DC source supplies energy to the inverter, which further energizes the three windings of the motor. The electrical signals from the motor are transformed into the stationary α-β reference frame, where the observer evaluates the state and generates the corrective signals corresponding to the back-EMF.

The back EMF that the observer estimates, with other derived quantities gives the sensorless drive the information it needs for rotor position and speed estimation. The commutation logic then chooses the inverter switching sequence, all without the use of Hall sensors, for position feedback.

The estimated rotor information is used to obtain the required switching states of inverter. Hall signals may be retained as reference signals for validating the estimated position during simulation, but they should not be used as the feedback source for the sensorless commutation path.

BLDC Motor Model for Observer Design

The observer is built on the electrical relationship between the stator voltage, phase current, and back-EMF. Assuming balanced phase windings with equal resistance and effective inductance, the phase-voltage equations are

$$v_a = R_s i_a + L_s \frac{di_a}{dt} + e_a, \tag{4.1}$$

$$v_b = R_s i_b + L_s \frac{di_b}{dt} + e_b, \tag{4.2}$$

$$v_c = R_s i_c + L_s \frac{di_c}{dt} + e_c, \tag{4.3}$$

where v_a, v_b, and v_c are the phase voltages; i_a, i_b, and i_c are the phase currents; e_a, e_b, and e_c are the phase back-EMFs; R_s is the phase resistance; and L_s is the effective phase inductance under the assumed model.

For a balanced three-wire motor,

$$i_a + i_b + i_c = 0. \tag{4.4}$$

The three-phase variables can therefore be transformed into two independent stationary-frame components using the Clarke transformation. This representation simplifies the mathematical implementation of the observer.

Clarke Transformation and State Variables

The phase currents are transformed into the α-β reference frame according to

$$\begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix}. \tag{4.5}$$

The same transformation is applied to the phase voltages and back-EMFs:

$$\begin{bmatrix} v_\alpha \\ v_\beta \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix}. \tag{4.6}$$

The transformed electrical model is expressed as

$$v_\alpha = R_s i_\alpha + L_s \frac{di_\alpha}{dt} + e_\alpha, \tag{4.7}$$

$$v_\beta = R_s i_\beta + L_s \frac{di_\beta}{dt} + e_\beta. \tag{4.8}$$

Rearranging these equations gives

$$\frac{di_\alpha}{dt} = -\frac{R_s}{L_s} i_\alpha + \frac{v_\alpha}{L_s} - \frac{e_\alpha}{L_s}, \tag{4.9}$$

$$\frac{di_\beta}{dt} = -\frac{R_s}{L_s} i_\beta + \frac{v_\beta}{L_s} - \frac{e_\beta}{L_s}. \tag{4.10}$$

These equations describe the current dynamics in the fixed reference frame. The current components i_α and i_β are the observer states, while the transformed voltages are model inputs. The back-EMF components are unknown quantities to be reconstructed by the observer. The measured currents are transformed and supplied to the observer. The currents estimated are then compared with the measured values to generate the sliding-mode correction signals.

4.3 Sliding Mode Observer Design

A SMO is a state estimator that uses an error dependent correction term to drive the difference, between measured and estimated states toward zero. In a SMO the signum function is often used to generate the switching correction. However the discontinuous nature of this function can introduce high-frequency oscillations known as chattering.

Qiao et al. [1] proposed replacing the signum switching function with a sigmoid function to reduce chattering in the estimated back-EMF. This principle is used as the basis of the observer investigated in this work.

Current Estimation Error

Let i_α and i_β denote the measured stationary-frame currents, and let $\hat{i}_\alpha$ and $\hat{i}_\beta$ denote their estimated values. The current estimation errors are defined as

$$\tilde{i}_\alpha = \hat{i}_\alpha - i_\alpha, \tag{4.11}$$

$$\tilde{i}_\beta = \hat{i}_\beta - i_\beta. \tag{4.12}$$

These errors indicate the difference between the observer states and the motor currents. The correction signals are generated from the errors and incorporated into the observer dynamics.

Sigmoid Switching Function

The conventional signum function can be given as

$$\operatorname{sgn}(S) = \begin{cases} +1, & S > 0, \\ 0, & S = 0, \\ -1, & S < 0, \end{cases} \tag{4.13}$$

where S is the sliding variable. Because this function changes abruptly at zero, its use can produce high-frequency switching in the observer correction signal.

To obtain a smoother transition, the sigmoid function can be expressed in the form

$$F(S) = \frac{2}{1 + \exp(-aS)} - 1, \tag{4.14}$$

which is equivalently written as

$$F(S) = \tanh\left(\frac{aS}{2}\right). \tag{4.15}$$

Here, a is a positive parameter that controls the slope of sigmoid function. For small values of S, the function changes smoothly around zero. For large positive or negative values of S, its output approaches $+1$ or -1, respectively.

The sigmoid function is continuous and bounded. Compared with an ideal signum function, it reduces the abrupt change in the correction signal near the sliding surface. The value of a affects the trade-off between smoothness and the sharpness of the switching response.

Observer State Equations

A representative current observer in the α-β reference frame can be expressed as

$$\frac{d\hat{i}_\alpha}{dt} = -\frac{R_s}{L_s}\hat{i}_\alpha + \frac{v_\alpha}{L_s} - \frac{z_\alpha}{L_s}, \tag{4.16}$$

$$\frac{d\hat{i}_\beta}{dt} = -\frac{R_s}{L_s}\hat{i}_\beta + \frac{v_\beta}{L_s} - \frac{z_\beta}{L_s}, \tag{4.17}$$

where z_α and z_β are the observer correction signals. The signs and scaling of these signals depend on the definitions adopted in the observer model.

Using the sigmoid switching function, a general correction structure is

$$z_\alpha = K_s F(\tilde{i}_\alpha), \tag{4.18}$$

$$z_\beta = K_s F(\tilde{i}_\beta), \tag{4.19}$$

where K_s is the observer gain $F(\cdot)$ is the sigmoid function defined in Equation (4.14).

The switching gain determines the strength of the correction applied to the estimated states. If the gain is too small, the estimation error may converge slowly or remain significant. If it is too large, the observer may become more sensitive to noise

and parameter mismatch.

4.4 Rotor Position and Speed Estimation

The purpose of the observer is to estimate electrical quantities that can be used to obtain information about rotor position and speed without using position sensors. In the proposed system, the observer correction signals are processed to estimate the back EMF components. These estimates are then used in the rotor information estimation subsystem.

The link between the estimated back-EMF, the rotor position and the speed depends on the motor's back-EMF waveform and on the estimation method being used. This is especially important for a BLDC motor because the phase back-EMFs in a BLDC motor are trapezoidal, than purely sinusoidal.

4.4.1 Back-EMF Estimation

Back-EMF estimation is an important step in sensorless BLDC motor control because it provides information about rotor motion. In this work, a sigmoid-based sliding mode observer (SMO) is used to estimate the back-EMF, following the method proposed by Qiao et al. [1]

The BLDC motor equations in the stationary α-β reference frame are

$$\frac{di_\alpha}{dt} = -\frac{R_s}{L_s} i_\alpha + \frac{v_\alpha}{L_s} - \frac{e_\alpha}{L_s}, \tag{4.20}$$

$$\frac{di_\beta}{dt} = -\frac{R_s}{L_s} i_\beta + \frac{v_\beta}{L_s} - \frac{e_\beta}{L_s}. \tag{4.21}$$

Here, i_α and i_β are the stator currents, v_α and v_β are the stator voltages, and e_α and e_β are the back-EMF components.

The current estimation errors are defined as

$$S_\alpha = \hat{i}_\alpha - i_\alpha, \qquad S_\beta = \hat{i}_\beta - i_\beta. \tag{4.22}$$

To reduce chattering caused by the conventional signum function, the observer uses the sigmoid switching function

$$F(S) = \frac{2}{1 + \exp(-aS)} - 1, \tag{4.23}$$

where a controls the slope of sigmoid function. The observer correction signals are

$$z_\alpha = K_s F(S_\alpha), \qquad z_\beta = K_s F(S_\beta), \tag{4.24}$$

where K_s is the switching gain.

The estimated currents are obtained using the observer equations

$$\frac{d\hat{i}_\alpha}{dt} = -\frac{R_s}{L_s}\hat{i}_\alpha + \frac{v_\alpha}{L_s} - \frac{z_\alpha}{L_s}, \tag{4.25}$$

$$\frac{d\hat{i}_\beta}{dt} = -\frac{R_s}{L_s}\hat{i}_\beta + \frac{v_\beta}{L_s} - \frac{z_\beta}{L_s}. \tag{4.26}$$

Under this observer's sign and scaling convention, the back-EMF estimates are

$$\hat{e}_\alpha = z_\alpha, \qquad \hat{e}_\beta = z_\beta. \tag{4.27}$$

In MATLAB/Simulink, the measured currents and the estimated currents are compared. The errors between measured currents and estimated currents are fed into a sigmoid function. The output of the sigmoid function is used to generate the observer correction signals. Then the back EMF estimated is compared with the reference that comes from the motor model. This comparison shows how well the estimation works and whether there is any chattering.

4.4.2 Rotor Position Estimation

The estimated back-EMF components contain information about the rotor's electrical position. The estimated electrical position can be obtained from the two back-EMF components as

$$\hat{\theta}_e = \text{atan2}(-\hat{e}_\alpha, \hat{e}_\beta). \tag{4.28}$$

Here, $\hat{e}_\alpha$ and $\hat{e}_\beta$ are the estimated back-EMF components, and $\hat{\theta}_e$ is the estimated electrical position. The signs and angle offset depend on the adopted reference-frame and back-EMF conventions and must be verified against the motor model.

The position estimated is compared with the actual rotor position available in MATLAB/Simulink to evaluate the accuracy of the sensorless estimation.

4.4.3 Rotor Speed Estimation

The back-EMF dynamics are described by the following equations, assuming that the electrical speed changes slowly over the observer time scale:

$$\dot{e}_\alpha = -\omega_e e_\beta, \qquad \dot{e}_\beta = \omega_e e_\alpha. \tag{4.29}$$

The back-EMF estimation errors are defined as

$$\tilde{e}_\alpha = \hat{e}_\alpha - e_\alpha, \qquad \tilde{e}_\beta = \hat{e}_\beta - e_\beta. \tag{4.30}$$

Using these errors, the back-EMF observer is expressed as

$$\dot{\hat{e}}_\alpha = -\hat{\omega}_e \hat{e}_\beta - l_2 \tilde{e}_\alpha, \tag{4.31}$$

$$\dot{\hat{e}}_\beta = \hat{\omega}_e \hat{e}_\alpha - l_2 \tilde{e}_\beta. \tag{4.32}$$

The rotor speed estimation error dynamics are given by

$$\frac{d\tilde{\omega}_r}{dt} = \tilde{e}_\alpha \hat{e}_\beta - \tilde{e}_\beta \hat{e}_\alpha. \tag{4.33}$$

Here, $\hat{\omega}_e$ is the estimated electrical speed, l_2 is the back-EMF observer gain, and $\tilde{\omega}_r$ is the rotor speed estimation error. The third equation updates the speed estimation error using the back-EMF estimation errors and estimated back-EMF components. Thus, speed estimation is coupled with the back-EMF observer.

In MATLAB/Simulink, the observer equations are implemented to estimate the back-EMF and rotor speed. The estimated speed is compared with the actual motor speed to evaluate the tracking performance of the observer.

4.5 Sensorless Control Implementation in MATLAB/Simulink

Table 4.1: BLDC Motor Parameters

Parameter	Symbol	Value
DC-link voltage	V_{dc}	120 V
Stator phase resistance	R_s	1.5 Ω
Stator phase inductance	L_s	0.5 mH
Voltage constant	K_e	70 V peak L-L/krpm
Back-EMF flat area	–	120°
Number of pole pairs	p	12
Moment of inertia	J	0.008 kg·m^2
Viscous damping coefficient	F	0.001 N·m·s
Static friction torque	T_f	0 N·m
Reference speed	N_{ref}	1000 rpm

4.5.1 Simulation Circuit Diagram

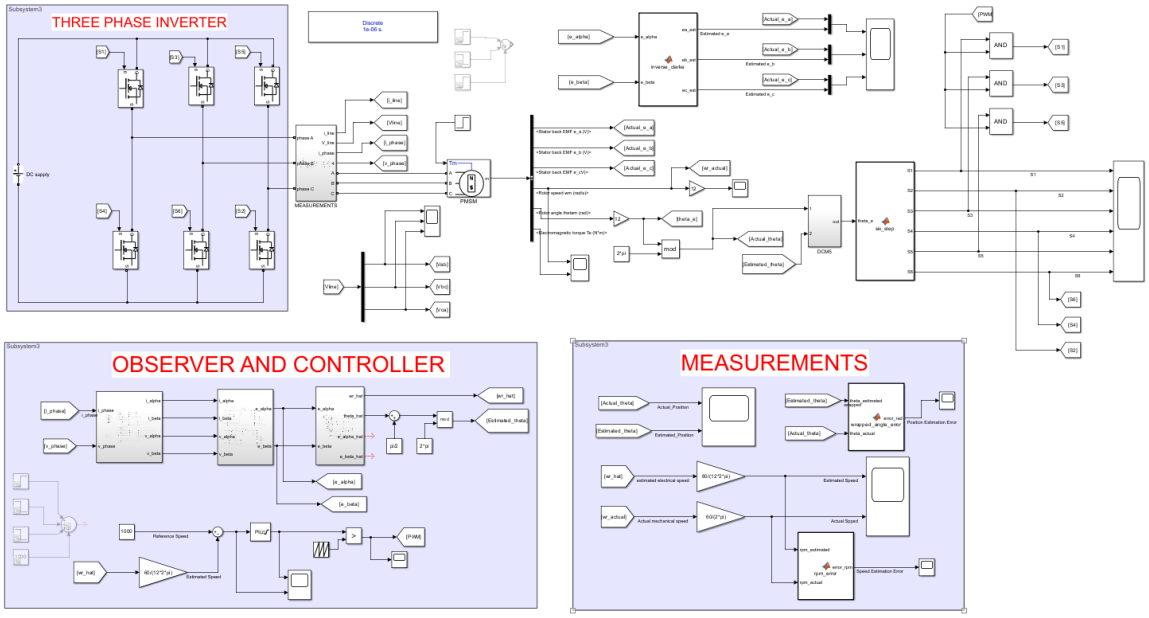


Figure 4.1: Complete Simulated Circuit Diagram of SMO based BLDC Drive Contol

4.6 Simulation Results

4.6.1 Back EMF

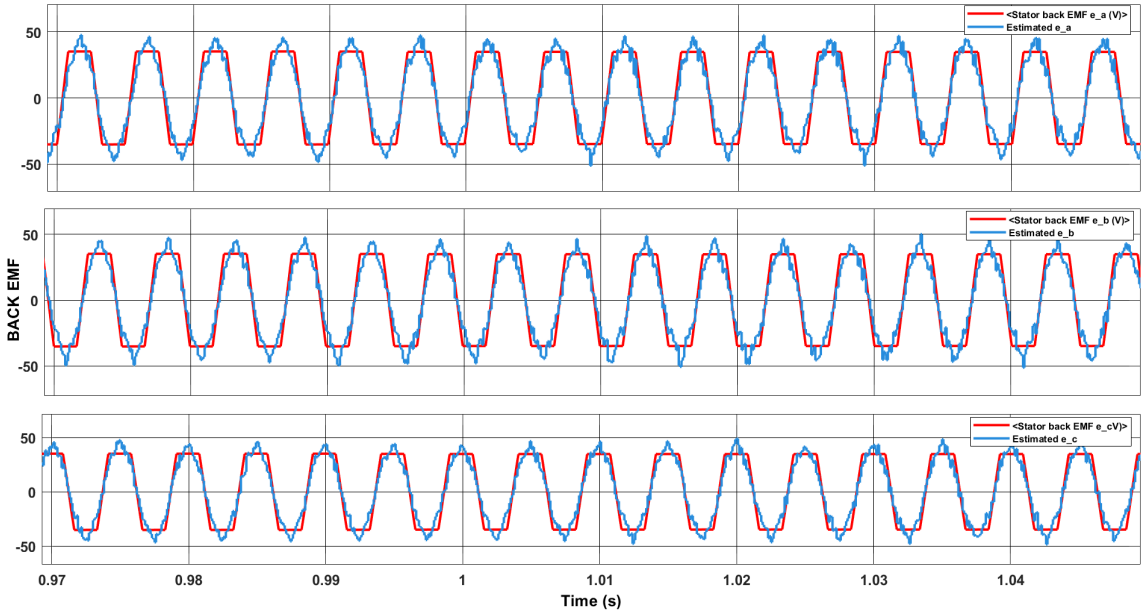


Figure 4.2: Actual and Estimated Back EMF

4.6.2 Rotor Position

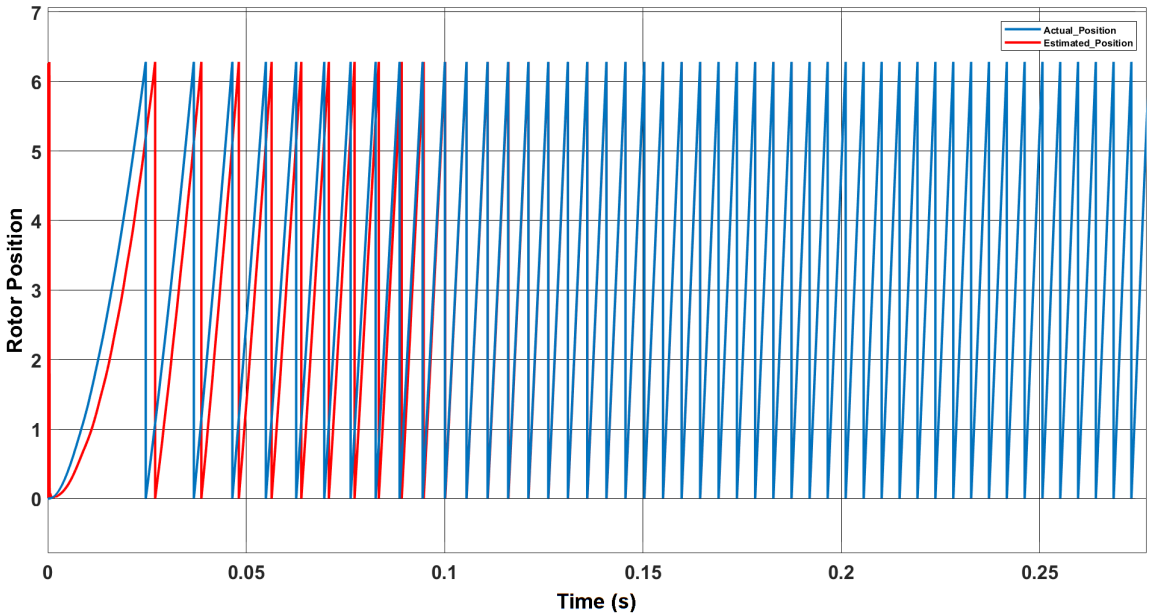


Figure 4.3: Actual and Estimated Rotor Position

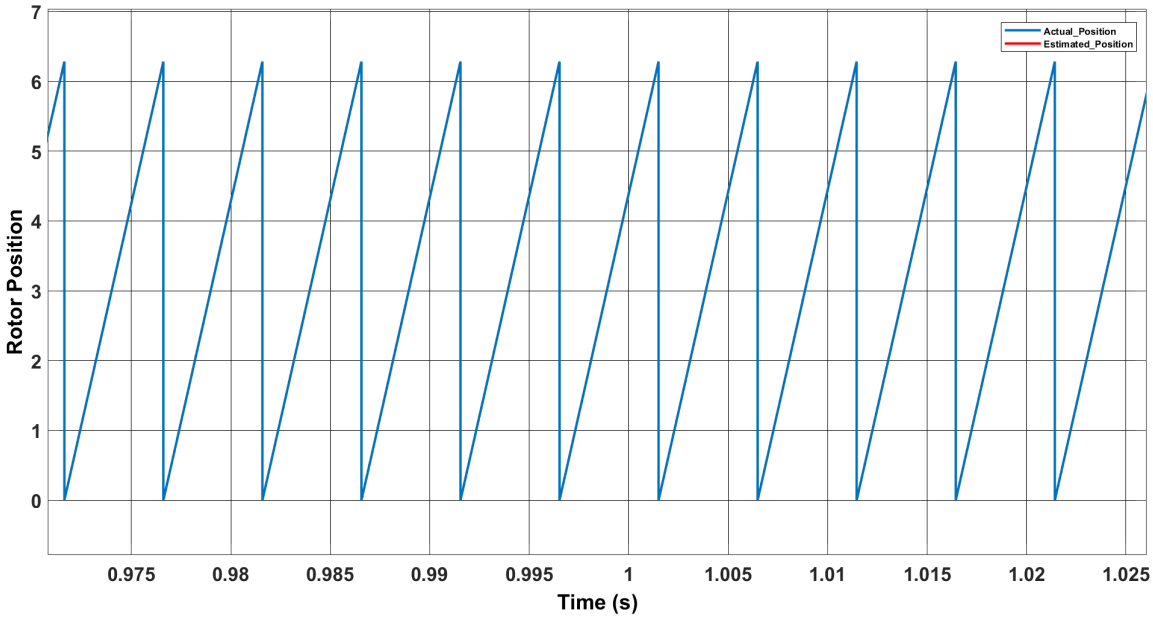


Figure 4.4: Zoomed View of Actual and Estimated Rotor Position

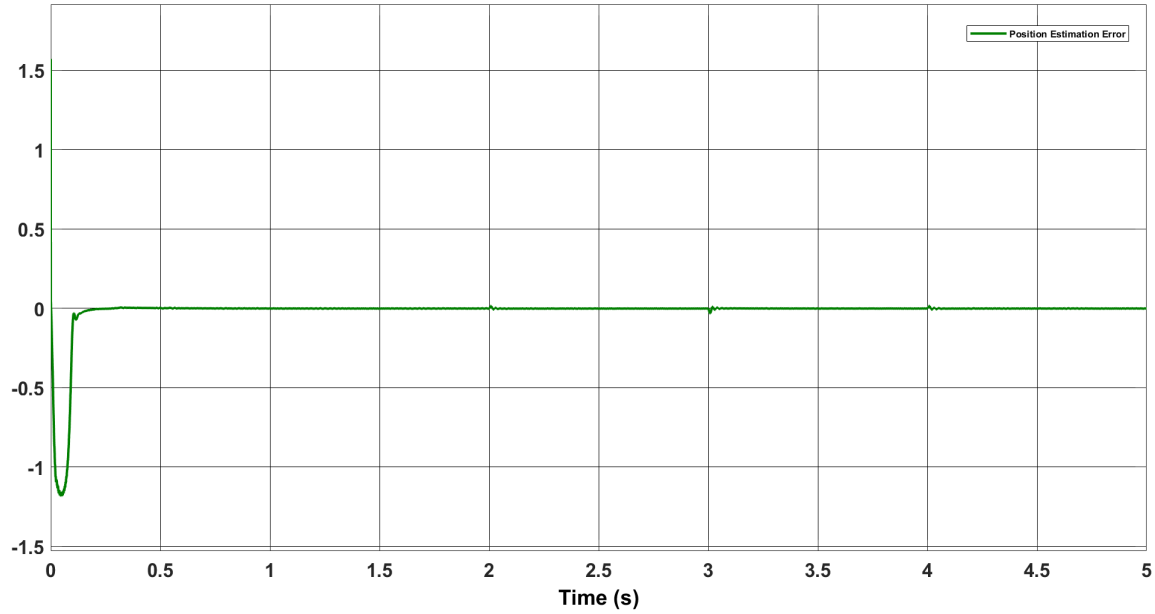


Figure 4.5: Rotor Position Estimation Error

4.6.3 Constant Speed, Variable Load Torque

To evaluate the disturbance rejection capability of the proposed approach, the motor is operated at a constant reference speed of 1000 RPM with an initial load torque of 0 Nm. The load torque is increased to 1 Nm at 2 s. It is reduced by 2 Nm bringing it to -1 Nm at 3 s, and it is increased by 1 Nm to return to 0 Nm at 4 s.

An increase in load torque tends to reduce motor speed, while a decrease in load torque reduces the mechanical loading. The speed controller adjusts its control input

to maintain the reference speed under these disturbances.

Aactual and estimated rotor speeds, along with the speed error, are examined to evaluate the observer's performance. The three-phase currents are also analysed to observe their response to changes in load torque.

Rotor Speed

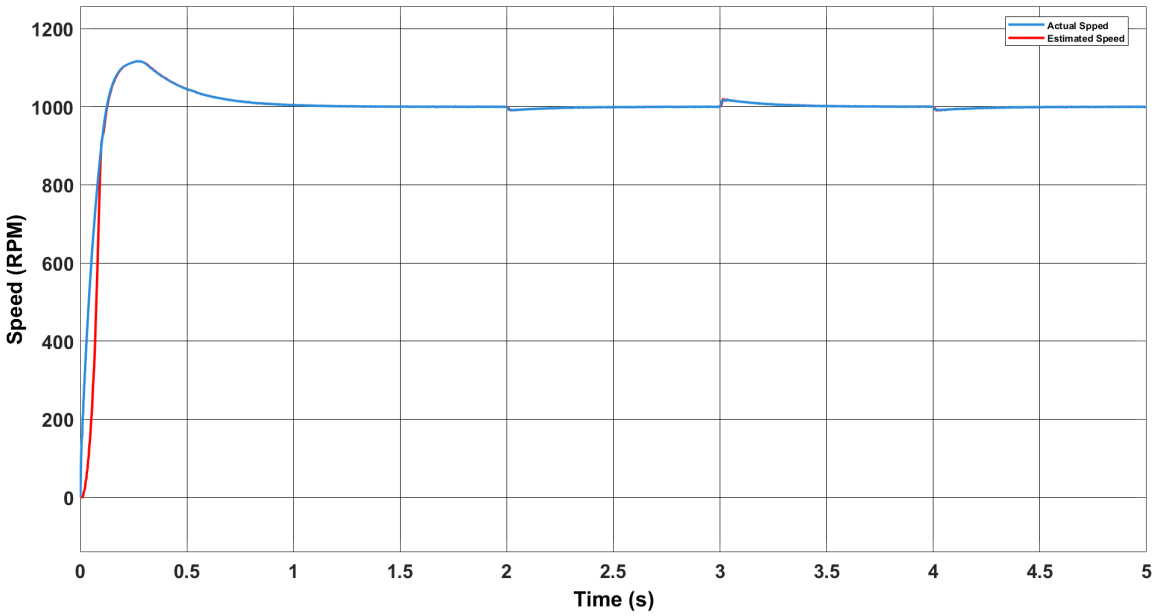


Figure 4.6: Actual and Estimated Rotor Speed under varying load torque

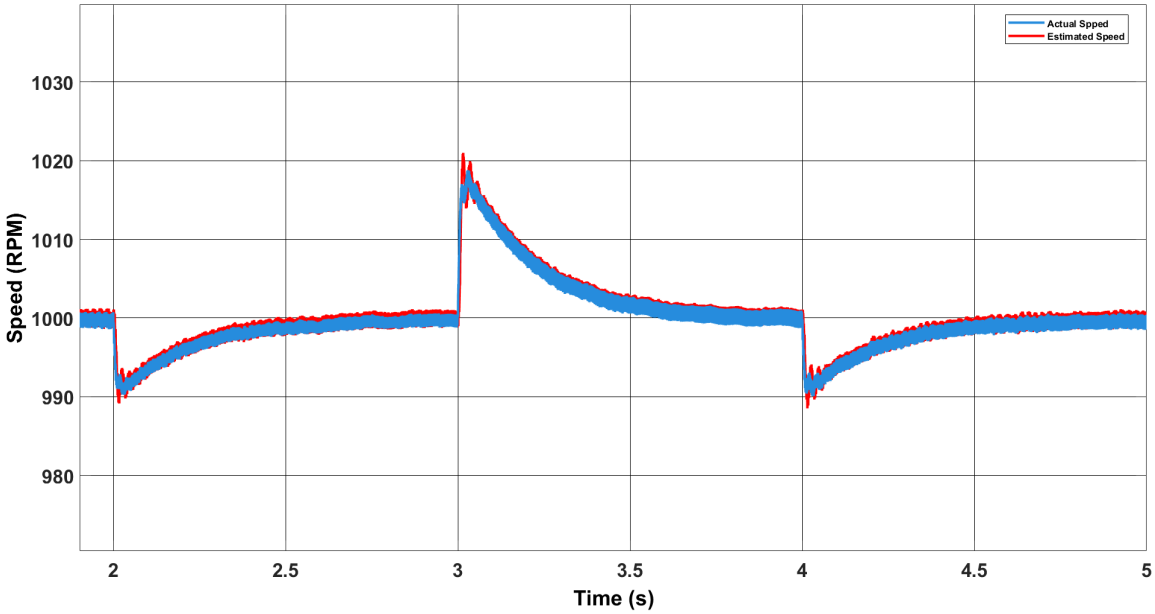


Figure 4.7: Zoomed View of Actual and Estimated Rotor Speed under varying load torque

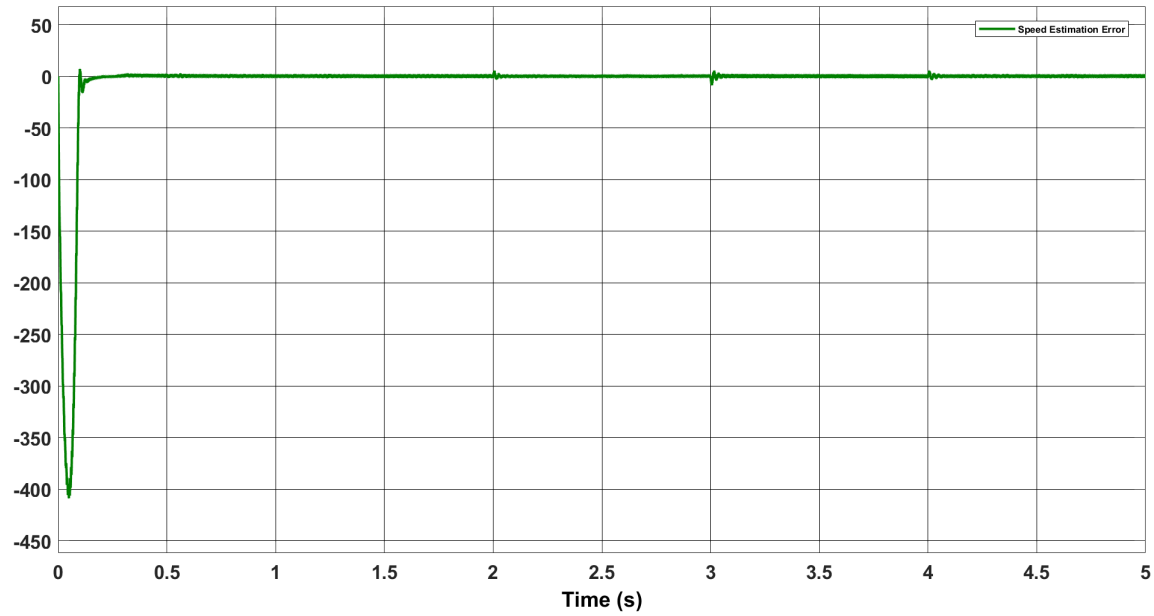


Figure 4.8: Rotor Speed Estimation Error under varying load torque

Three-Phase Currents

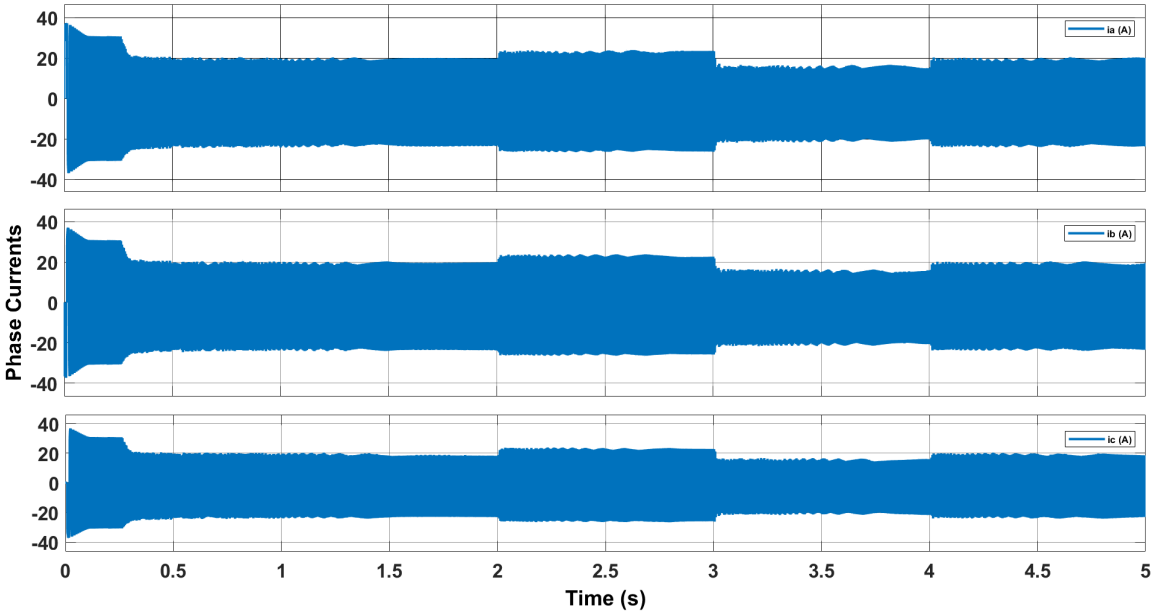


Figure 4.9: Three-Phase Currents under varying load torque

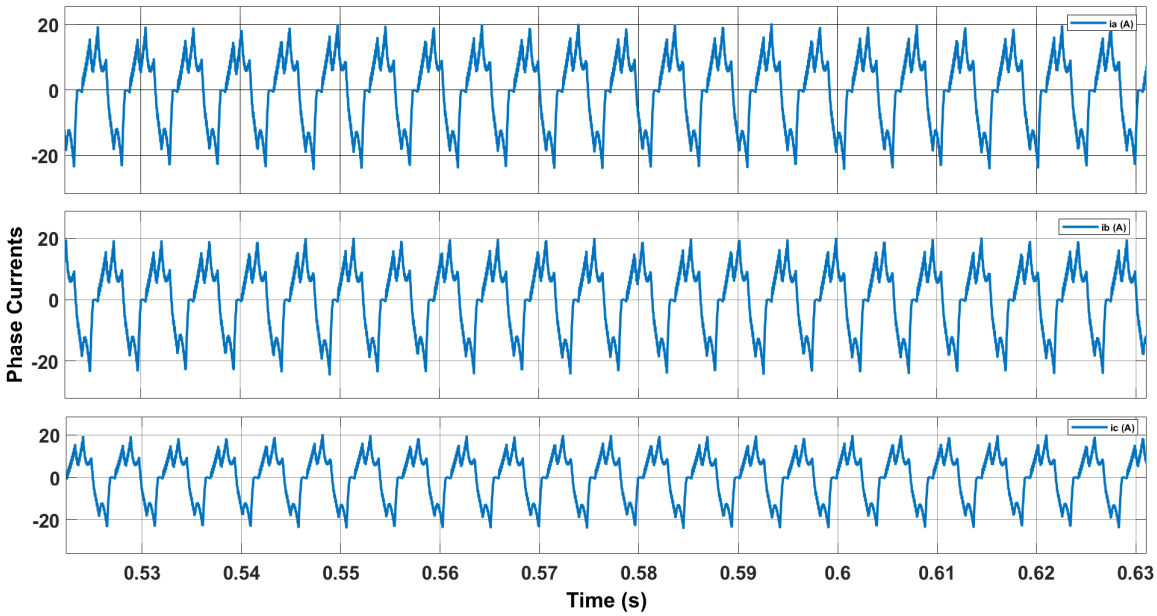


Figure 4.10: Zoomed View of Three-Phase Currents under varying load torque

4.6.4 Constant Load Torque, Variable Speed

To evaluate the speed-tracking performance of the proposed sensorless BLDC drive, the motor is operated at a constant load torque of 1 Nm while the reference speed is varied. Initially, the reference speed is set to 1000 RPM. At 1.5 s, it is reduced by 50 RPM to 950 RPM, followed by a further reduction of 50 RPM to 900 RPM at 2 s. Finally, at 2.5 s, the reference speed is increased by 100 RPM to return to initial reference of 1000 RPM.

The speed controller adjusts its control input according to the changes in reference speed to achieve the desired operating points. Actual and estimated rotor speeds, along with the speed estimation error, are examined to evaluate the tracking performance of the proposed observer. The three-phase currents are also analysed to observe their response during motor acceleration and deceleration.

Rotor Speed

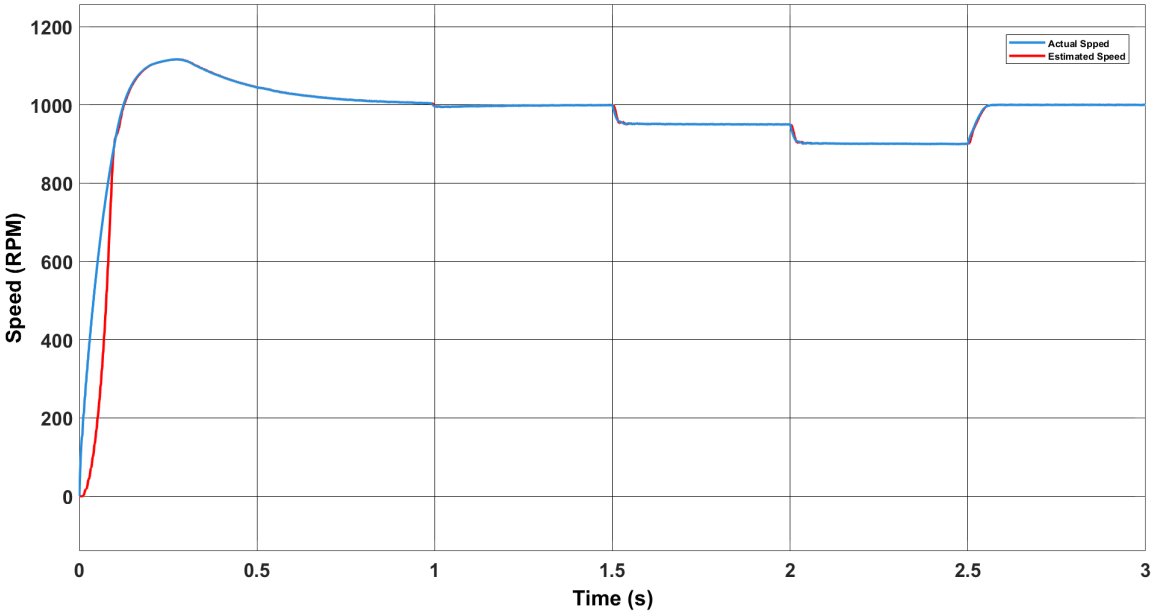


Figure 4.11: Actual and Estimated Rotor Speed under varying speed

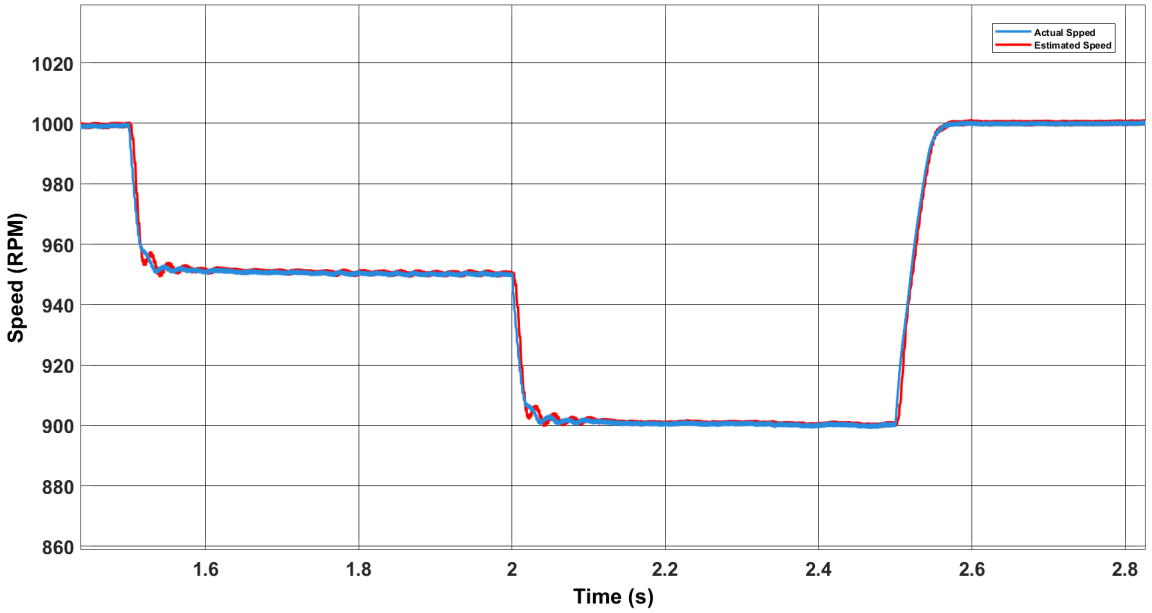


Figure 4.12: Zoomed View of Actual and Estimated Rotor Speed under varying speed

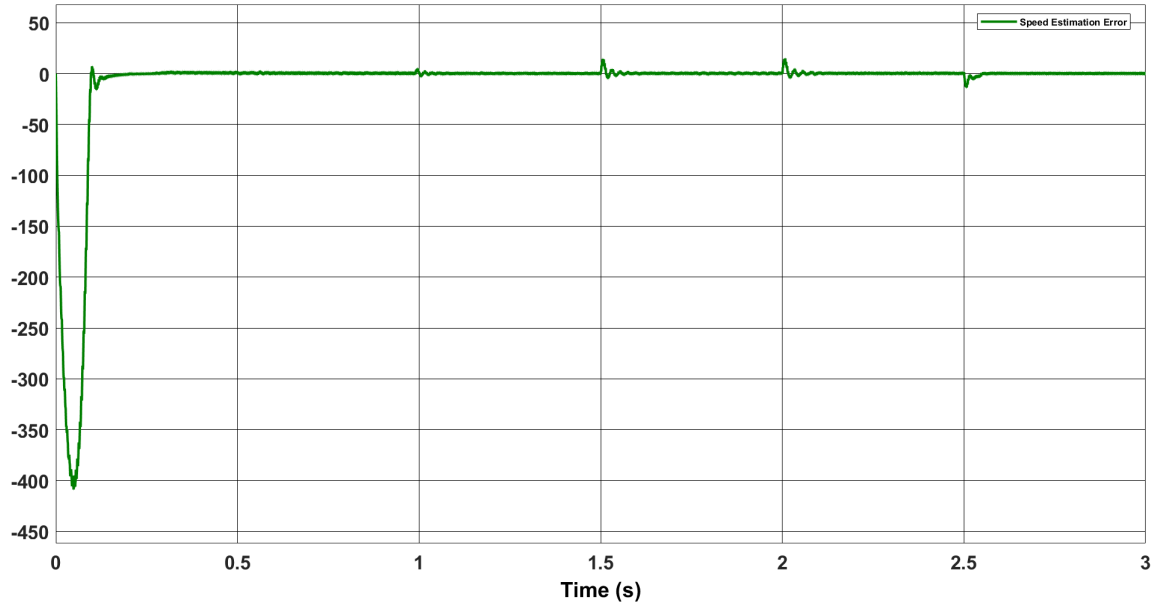


Figure 4.13: Rotor Speed Estimation Error under varying speed

Three-Phase Currents

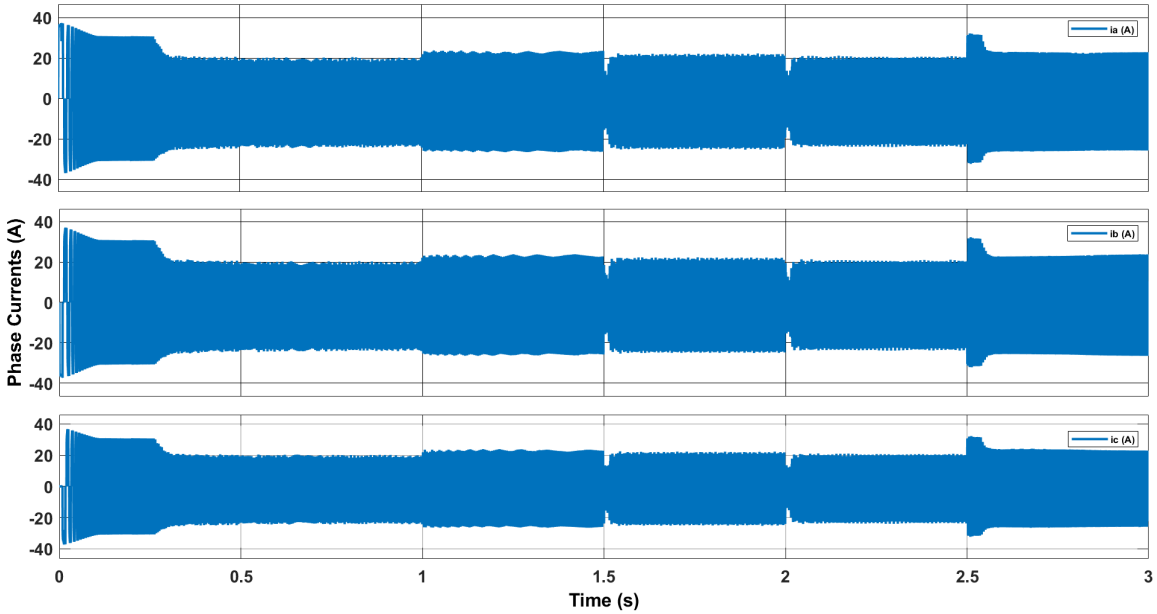


Figure 4.14: Three-Phase Currents under varying speed

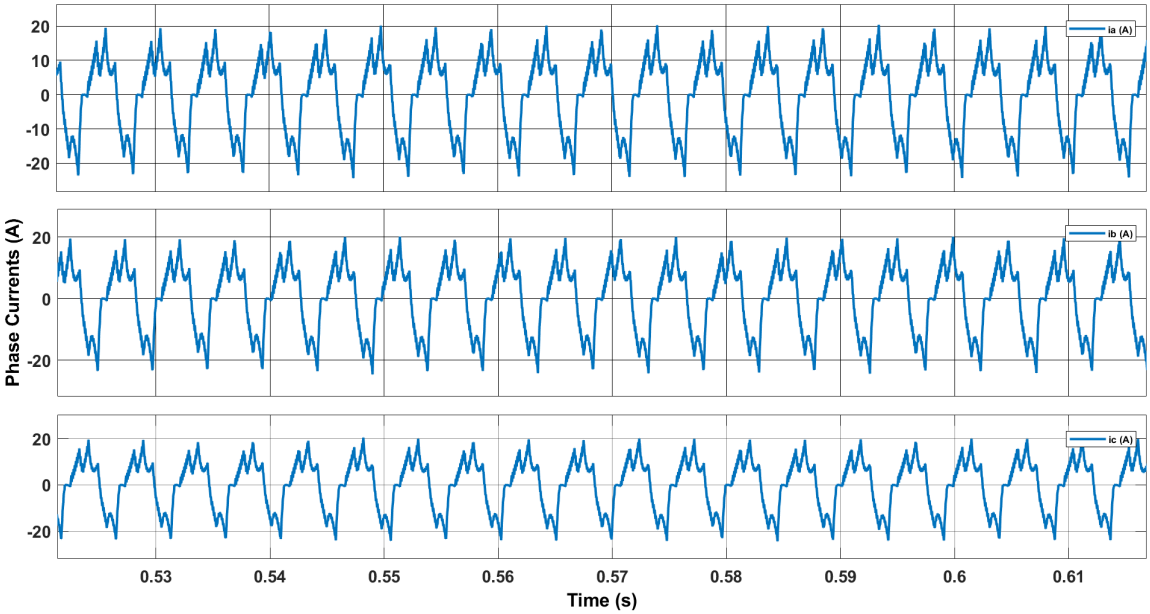


Figure 4.15: Zoomed View of Three-Phase Currents under varying speed

4.7 Limitations of the Present Simulation Study

The present work investigates sensorless control of a BLDC motor using a Sliding Mode Observer (SMO) implemented in MATLAB/Simulink. Although the simulation results demonstrate the ability of the observer to estimate back-EMF and rotor position, certain limitations remain.

1. **Dependence on Motor Parameters:** The observer relies on the mathematical model of the BLDC motor. Variations in stator resistance, inductance, and other motor parameters may affect the accuracy of the estimated signals.

2. **Approximation Using Sigmoid Function:** Instead of the conventional signum function a sigmoid function is used in the sliding mode observer to obtain a smoother switching action. Although this helps reduce chattering, the approximation may introduce estimation errors depending on the sigmoid slope and selected parameters. Proper tuning is therefore required to maintain a balance between estimation accuracy and smoothness.

3. **Low-Speed Operation:** The back-EMF magnitude decreases with motor speed. Consequently, accurate estimation becomes difficult at very low speeds, particularly during startup and operation near standstill.

4. **Simulation-Based Validation:** The proposed method has been evaluated using MATLAB/Simulink. Practical effects such as measurement noise, inverter

dead time, semiconductor voltage drops, and hardware imperfections have not been fully considered. These effects may influence observer performance during experimental implementation.

5. **Observer Parameter Selection:** The performance of the SMO depends on the selection of observer gains and sigmoid function parameters. Improper tuning may result in slower convergence, increased estimation error, or oscillations in the estimated signals.

These limitations indicate that further investigation is required before applying the developed observer to a practical BLDC motor drive. Experimental validation, improved parameter tuning, and evaluation under different operating conditions can be considered for future work.

Chapter 5

Conclusion and Future Work

5.1 Summary of Work Completed

The present work focused on the development and simulation of a sensorless control scheme for a BLDC motor using a sliding mode observer. The main objective was to estimate rotor position and speed without using physical rotor position sensors and to use these estimated quantities for controlling the motor.

The work was initially inspired by the sliding mode observer proposed by Qiao et al. [1]. However, the original method was developed for a Permanent Magnet Synchronous Motor (PMSM) and used field-oriented control (FOC) with space vector PWM. In the present work, the observer approach was adapted for a BLDC motor, and the control structure was modified to use conventional six-step commutation.

The mathematical model of the BLDC motor was developed using its electrical and mechanical equations. The sliding mode observer was implemented using the motor model to estimate the back-EMF. The conventional signum function was replaced with a sigmoid function to obtain a smoother switching action and reduce chattering. Using the estimated back-EMF, the rotor position and speed were obtained through the observer structure.

After estimating the rotor speed, a PI controller is implemented to regulate the motor speed. The difference between the reference speed and estimated speed was given as the input to the PI controller. Its output was used to generate the PWM command for controlling the inverter.

The estimated rotor position was supplied to the six-step commutation logic to obtain the appropriate switching logic. The PWM command was combined with the commutation logic to generate the required inverter gate pulses. Thus, the FOC and space vector modulation structure used in the original paper was not retained. Instead, the proposed system uses a speed PI controller and six-step electronic com-

mutation for BLDC motor operation.

The complete system was implemented in MATLAB/Simulink. The simulation results were examined to study the estimated back-EMF, rotor position, rotor speed, and motor response under the different operating conditions.

5.2 Conclusions from Simulation Studies

The simulation results offer an insight into the performance of the proposed sensorless control strategy for the SMO based BLDC motor. The conclusions that can be made from the study are as follows.

1. Mathematical modeling of BLDC motor was done which served as a foundation for constructing the proposed scheme.
2. The sliding mode observer technique was derived on the basis of the method described in the paper by Qiao et al.[1] and was applied to the BLDC motor.
3. Instead of the usual signum function, the sigmoid function was used in the observer. It provides smooth transition of states of the observer and reduces the effect of chattering in the sliding mode.
4. The observer was used to estimate the back-EMF, rotor position, and rotor speed. The results were compared to the real values.
    (a) The PI controller was applied to regulate the motor speed. The estimation of rotor speed was compared with reference speed, and accordingly, the controller output was calculated for determination of the PWM command signal.
    (b) The estimated value of the rotor position was applied to the six-step commutation algorithm for selection of the inverter switching states. The output commutation signals were superimposed with the PWM command for generation of inverter gate pulses.

Thus, the simulation research shows that the control system with an observer-based approach to the sensorless control of a BLDC motor drive has been implemented. It estimates the rotor position and speed for commutation and speed regulation purposes. It also allows one to study the behavior of the sigmoid-based sliding mode observer.

5.3 Limitations of the Present Work

Though the above results from simulation analysis give insights into the proposed sensorless control strategy, there are still some limitations.

1. **Dependency on Motor Characteristics:** The observer is designed on the basis of mathematical model of BLDC motor. Any variations in stator resistance, inductance and other characteristics of the motor will affect the performance of the signal estimation.

2. **Estimation Using Sigmoid Function:** The use of sigmoid function gives more accurate switching actions than the traditional signum function. But the accuracy of estimation depends on the chosen slope parameter.

3. **Operation at Low Speed:** The magnitude of Back EMF decreases with the increase in speed of the motor. At very low speed operation, especially in standstill position, the estimation becomes inaccurate.

4. **Validation Based on Simulations:** The proposed control strategy has been validated through simulations using MATLAB/Simulink. Effects such as measurement errors, dead time of inverter, voltage drop across the semiconductors are yet to be explored.

5. **Tuning of the Controller and Observer:** The performance of the complete system depends on the design of observer gains, sigmoid function parameter and PI controller gains.

These limitations indicate that further investigation is required before applying the proposed control scheme to a practical BLDC motor drive.

5.4 Proposed Hardware Implementation

Next step toward practical implementation of the sensorless control scheme would be its realization via hardware based BLDC motor drive. In particular, the system to be realized would include a BLDC motor, three phase voltage source inverter, DC supply, digital controller, and associated electrical measurement circuits.

Obtained measurements of electrical values will serve as input data for the sliding mode observer that will calculate the back EMF, rotor position, and rotor speed values. The estimated value of rotor speed will be compared with the reference speed value and PI controller will form the required PWM command.

The position of rotor that is estimated will be used in the six-step commutation logic to calculate the conducting inverter switches. The PWM command and commutation signals will be used for formation of gate pulse signals for the three-phase inverter.

An appropriate digital controller (DSP or microcontroller) may be utilized for realization of the observer, PI controller, and commutation logic. Realization of the scheme in hardware will allow studying the effects associated with measurement noises, switching delays, parameter changes, etc., that cannot be accurately simulated in the computer model.

5.5 Future Scope

Some of the ways to further enhance the performance of the present sensorless BLDC motor control technique are as follows:

1. **Hardware Implementation:** The observer design as well as the six-step control strategy can be realized using a DSP or a microcontroller to test the performance on a BLDC motor drive.

2. **Performance Enhancement of Observer:** Some more studies can be carried out to tune the observer gains and parameters for the sigmoid function to enhance the estimation performance.

3. **Sensorless Operation at Low Speeds:** More estimation methods can be studied to enhance the performance of position estimation at low speeds and at start up when back-EMF magnitude is low.

4. **Speed Control by PI Controller:** Some more methods can be adopted to tune the PI controller for improved performance at changing loads.

5. **Robustness Study of Scheme:** The performance of the proposed scheme can be examined for different motor parameters, varying load conditions and disturbance in measurements.

6. **Real Time Performance Assessment:** Real time performance analysis of the observer, PI controller and six-step commutation logic can be carried out.

These improvements can contribute to extending the current simulation study into a realistic implementation of a sensorless BLDC motor control with sliding mode observer-based rotor position and speed estimation.

Bibliography

[1] Z. Qiao, T. Shi, Y. Wang, Y. Yan, C. Xia, and X. He, "New Sliding-Mode Observer for Position Sensorless Control of Permanent-Magnet Synchronous Motor," *IEEE Transactions on Industrial Electronics*, vol. 60, no. 2, pp. 710–719, Feb. 2013, doi: 10.1109/TIE.2012.2206359.

[2] J. C. Gamazo-Real, E. Vázquez-Sánchez, and J. Gómez-Gil, "Position and Speed Control of Brushless DC Motors Using Sensorless Techniques and Application Trends," *Sensors*, vol. 10, no. 7, pp. 6901–6947, 2010, doi: 10.3390/s100706901.

[3] H. Fakham, M. Djemai, and K. Busawon, "Design and Practical Implementation of a Back-EMF Sliding-Mode Observer for a Brushless DC Motor," *IET Electric Power Applications*, vol. 2, no. 6, pp. 353–361, 2008, doi: 10.1049/iet-epa:20070242.

[4] O. G. Çetintaş, K. Akgül, and L. T. Ergene, "Position Sensorless Speed Control of BLDC Motor with Using Back-EMF Method," in *Proceedings of the 2023 International Conference on Electrical and Electronics Engineering (ELECO)*, 2023, doi: 10.1109/ELECO60389.2023.10415959.

[5] H. Fakham, M. Djemai, A. Reama, and P. Blazevic, "Sliding Mode Observer for Position and Speed Estimations in Brushless DC Motor (BLDCM)," in *Proceedings of the 2004 IEEE International Conference on Industrial Technology (ICIT)*, vol. 1, pp. 121–126, 2004.

[6] A. Deenadayalan, C. Dhananjai, and G. Saravana Ilango, "Modified Sliding Mode Observer for Wide Speed Range Operation of Brushless DC Motor," *Frontiers of Electrical and Electronic Engineering*, vol. 7, no. 4, pp. 467–476, 2012, doi: 10.1007/s11460-012-0210-2.

[7] M.-G. Gan, M. Zhang, C.-Y. Zheng, and J. Chen, "An Adaptive Sliding Mode Observer Over Wide Speed Range for Sensorless Control of a Brushless DC Motor," *Control Engineering Practice*, vol. 77, pp. 52–62, Aug. 2018, doi: 10.1016/j.conengprac.2018.05.004.

[8] Q. Liang and Z. Zhao, "A Position Sensorless Control Strategy for BLDCM Based on a Fractional Order Sliding Mode Observer," *Journal of Shandong University (Engineering Science)*, vol. 47, no. 3, pp. 96–101, 2017, doi: 10.6040/j.issn.1672-3961.0.2016.298.

[9] L. Qiu, M. Qu, and X. Liu, "Position Sensorless Control of Brushless DC Motor Based on Sliding Mode Observer," in *Proceedings of the 5th EAI International Conference on Advanced Hybrid Information Processing (ADHIP 2021)*, Part II, Lecture Notes of the Institute for Computer Sciences, Social Informatics and Telecommunications Engineering, vol. 417, pp. 342–355, 2022, doi: 10.1007/978-3-030-94554-1_27.

[10] R. Zhang, G. Bao, B. Wang, and Y. Zhang, "BLDCM Position Sensorless Control Based on Variable Reaching Law Sliding Mode Back EMF Observer," *Electric Machines & Control Application*, vol. 49, no. 8, pp. 1–6, 2022, doi: 10.12177/emca.2022.071.

[11] A. Darba, F. De Belie, and J. Melkebeek, "Sensorless Commutation and Speed Control of Brushless DC-Machine Drives Based on the Back-EMF Symmetric Threshold-Tracking," in *Proceedings of the 9th IEEE International Conference on Electrical Machines and Drives (IEMDC)*, pp. 492–497, 2013, doi: 10.1109/IEMDC.2013.6556141.

[12] V. Aishwarya and B. Jayanand, "Estimation and Control of Sensorless Brushless DC Motor Drive Using Extended Kalman Filter," in *Proceedings of the International Conference on Circuit, Power and Computing Technologies (ICCPCT)*, 2016, doi: 10.1109/ICCPCT.2016.7530343.

[13] U. K. Soni and R. K. Tripathi, "Back EMF Noise Band Based Initial Startup Scheme for Sensorless Operation of BLDC Drive," in *Proceedings of the 2019 Innovations in Power and Advanced Computing Technologies (i-PACT)*, Vellore, India, 2019, doi: 10.1109/i-PACT44901.2019.8960196.

[14] H. Zhang, G. Liu, X. Zhou, and S. Zheng, "High-Precision Sensorless Optimal Commutation Deviation Correction Strategy of BLDC Motor With Asymmetric Back EMF," *IEEE Transactions on Industrial Informatics*, vol. 17, no. 8, pp. 5250–5259, Aug. 2021, doi: 10.1109/TII.2020.3027010.